Understanding Multilateral Tariffs and Exchange Rates*

Jason Lu
International Monetary Fund
Research Department
jlu2@imf.org

Dimitre Milkov
International Monetary Fund
Research Department
dmilkov@imf.org

Abstract

The standard prediction that an import tariff appreciates the tariffing country's currency comes from a two-country setting, where the bilateral exchange rate, effective exchange rate, and the terms of trade coincide. In a multilateral setting, these measures separate and respond differently to tariffs. Under symmetric trade relationships, we show that a unilateral tariff appreciates the effective exchange rate but can depreciate the home currency against untariffed bystanders when substitution across foreign suppliers is sufficiently elastic. Tariff retaliation can instead lead both counterparts to depreciate in effective terms. These results are robust to moderate asymmetry, but under strong asymmetry, tariffs on small trading partners can worsen the tariffing country's terms of trade and reduce welfare when foreign suppliers are sufficiently close substitutes. We use the framework to contrast exchange-rate responses during the 2025 tariff episode.

Keywords: multilateral tariffs, exchange rates, trade diversion, terms of trade, Marshall–Lerner condition.
JEL codes: F13, F31, F41.

*Preliminary draft; this version October 29, 2026. The views expressed are those of the authors and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

1 Introduction

What happens to exchange rates when a country imposes a tariff? In the standard two-country model, a tariff shifts expenditure away from imports and improves the trade balance at the prevailing exchange rate. Under the Marshall–Lerner condition, external balance is restored through an appreciation of the tariffing country's currency (Dornbusch, 1974; Krugman, 1982). Because there is only one trading partner, the bilateral exchange rate is also the effective exchange rate. In the long run, in real terms, this same relative price also governs the terms of trade. Thus, in the simplest bilateral setting, the bilateral exchange rate, the effective exchange rate, and the long-run terms of trade all necessarily coincide.

These concepts separate once trade is multilateral. A tariff on one partner changes not only the relative price of foreign and domestic goods, but also relative prices across foreign suppliers. It can therefore affect the exchange rate against the target, the rates against untariffed bystanders, and the effective exchange rate differently. In any differentiable multi-country trade environment, we show that the first-order response can be represented as the adjustment of exchange rates to a vector of tariff-induced trade-balance disturbances. We define a multilateral Marshall–Lerner condition, implied by gross substitutability of trade balances in currency values, that ensures well-behaved adjustment. Crucially, we show that even under this condition, the sign of an individual bilateral exchange-rate response is not determined. A tariff therefore need not appreciate the tariffing country's currency against every partner.

We first answer this question in a general trade environment under symmetry among trading partners. We show that any vector of multilateral tariff changes decomposes into average protection and relative treatment. Average protection uniquely determines the change in the nominal effective exchange rate, while relative treatment determines the relative changes in bilateral exchange rates. Under a unilateral tariff, these two components reinforce each other against the targeted partner but work in opposite directions against untariffed bystanders. The response against bystanders is therefore ambiguous and depends on the structure of the trade environment.

To characterize these counteracting effects, we develop a nested CES Armington model that distinguishes substitution between domestic and foreign goods from substitution across foreign suppliers. Substitution toward domestic goods supports the conventional appreciation, while substitution across foreign suppliers shifts expenditure toward untariffed countries and strengthens their currencies relative to the tariffing country. Allowing these two margins to differ, we show that the tariffing country can depreciate against an untariffed bystander when cross-foreign substitution is sufficiently strong relative to home-foreign substitution,

but, under weak home bias, this outcome cannot arise with a single elasticity across domestic and foreign varieties. Empirical estimates distinguishing cross-foreign from home-foreign substitution support treating these margins separately (Broda and Weinstein, 2006; Feenstra et al., 2018; Li et al., 2022).

The same mechanisms extend naturally from unilateral tariffs to retaliation and multilateral trade wars. When the target retaliates with an equal tariff, the average-protection effects on the bilateral exchange rate cancel, while the relative-treatment effects reinforce each other against third countries. As a result, the threshold for depreciation against bystanders is substantially lower than under a unilateral tariff. In a multilateral trade war, in which every partner retaliates against the tariffing country, the same condition leads it to depreciate against every partner.

Our trade framework is deliberately stylized and is not intended to provide a quantitative account of the 2025 tariff episode, but it nonetheless offers a new perspective on the exchange-rate movements during this period. The dollar's depreciation following the tariff announcements was initially interpreted in part as evidence of higher risk premia, declining convenience yields, and concerns about the dollar's global role (Itskhoki and Mukhin, 2025; Kalemli-Özcan et al., 2026). Applying our framework to the broad features of the episode, we show that dollar depreciation need not originate outside the trade channel, particularly when retaliation is expected. Consistent with this explanation, Ostry et al. (2025) document that the dollar tends to appreciate following unilateral US tariffs, depreciate when trading partners retaliate, and remain comparatively stable against the renminbi during US–China tit-for-tat episodes. They hypothesize that dominant-currency pricing may help explain this pattern. Our results show that the same pattern can arise from canonical trade substitution and diversion effects once trade and exchange-rate adjustment is considered in a multilateral setting.

In a general asymmetric setting, where economies differ in size, trade shares, and productivities, exchange-rate responses no longer admit simple analytical expressions that characterize when one force dominates another. We show that the main qualitative results are robust to moderate asymmetry, although under strong asymmetry some aggregate conclusions can reverse. In particular, when cross-foreign substitution is very strong and the targeted country accounts for only a small share of the tariffing country's trade, diversion toward more important bystanders can dominate the average-protection effect and worsen the aggregate terms of trade. We examine these reversals numerically in the three-country model and find that they require relatively extreme configurations of country size and trade shares. This possibility is therefore less relevant for understanding the 2025 tariff episode, when the United States imposed substantial protection on its major trading partners.

These reversals apply to discriminatory tariffs, not to uniform protection. Under general conditions allowing country sizes, trade shares, home bias, and preference parameters to differ, we prove that a small uniform tariff imposed from a balanced free-trade baseline appreciates the tariffing country's currency weakly against every partner under the multilateral Marshall–Lerner condition. It therefore improves the aggregate terms of trade and raises welfare to first order. This result recovers the classical terms-of-trade argument for uniform protection while showing why its partner-specific counterpart can fail. Uniform protection eliminates relative treatment across suppliers and switches off the cross-origin diversion mechanism that drives the reversals.

Our paper builds on the classical analysis of tariffs under flexible exchange rates, which identifies the conditions under which tariffs lead to exchange-rate appreciation (Lerner, 1936; Metzler, 1949; Mundell, 1961; Dornbusch, 1974; Eichengreen, 1981; Krugman, 1982). A large open-economy macroeconomics literature extends this analysis to environments with nominal rigidities, monetary policy, currency invoicing, and alternative uses of tariff revenue (Lindé and Pescatori, 2019; Barattieri et al., 2021; Erceg et al., 2023; Bergin and Corsetti, 2023; Werning et al., 2025). We instead revisit the flexible-price benchmark in a multilateral setting, where tariffs affect not only the importing country's aggregate exchange rate but also relative exchange rates across trading partners. Our bystander results are related to the literature on third-country effects of preferential trade policy, particularly the roles of trade diversion and competition among exporters (Viner, 1950; Mundell, 1964; Bond and Syropoulos, 1996; Bagwell and Staiger, 1999, 2002). We show how these forces affect bilateral and effective exchange rates when the importing country also produces a competing good. Finally, our multilateral Marshall–Lerner condition draws on the classical literature on stability in multiple markets (Hicks, 1939; Mosak, 1944; Metzler, 1945; Arrow and Hurwicz, 1958), applying these conditions to the exchange-rate response to tariffs in a multilateral setting.

Our results also relate to the empirical literature on the exchange-rate effects of tariffs, including the 2025 episode (Itskhoki and Mukhin, 2025; Kalemli-Özcan et al., 2026; Ostry et al., 2025) and earlier episodes (Furceri et al., 2019; Matveev and Ruge-Murcia, 2021; Jeanne and Son, 2024). Our analysis also offers a complementary perspective on the empirical literature estimating tariff pass-through into import prices (Amiti et al., 2019; Fajgelbaum et al., 2020; Cavallo et al., 2021). This literature typically uses dollar-denominated import prices as a proxy for tariff incidence. In a multilateral setting, however, exchange-rate movements provide an additional margin of incidence through changes in relative prices. An exporter whose dollar price is unchanged may still experience a decline in the home-currency value of its receipts when the dollar depreciates. Assessing tariff incidence from import prices

alone can therefore overlook adjustment through exchange rates, a margin that may have been particularly relevant during the broad dollar depreciation in 2025.

This paper is organized as follows. Section 2 develops the general multilateral framework, derives the multilateral Marshall–Lerner condition, and decomposes tariff changes into average protection and relative treatment. Section 3 gives this decomposition economic content in a nested-CES model and derives the main results for unilateral tariffs. Section 4 extends the analysis to bilateral retaliation, multilateral trade wars, and preferential tariffs. Section 5 considers asymmetric trade relationships, examines how the results change away from symmetry, and applies the framework to the 2025 tariff episode. Section 6 concludes, with derivations, proofs, and additional robustness results provided in the appendices.

2 Tariffs and Exchange-Rate Adjustment

2.1 The bilateral benchmark

Consider two countries, A and B. Let e_{AB} denote units of A's currency per unit of B's, so that a rise is a depreciation of A, and write $\nu \equiv \log e_{AB}$. Country A levies an ad valorem tariff τ on imports from B.

Each country's trade balance TB_i is the value of its exports minus the value of its imports, both at border (pre-tariff) prices and both expressed in one fixed common unit of account. Tariff revenue is a domestic transfer: it is counted neither as export revenue nor as an import payment, although its rebate or other use may affect income, quantities and therefore the trade balance. In the nested-CES model of Section 3 it is rebated lump-sum. Let v_l denote the common-unit value of one unit of currency l. We select A as numeraire and normalize $v_A = 1$, so these trade-balance values are numerically equal to values expressed in A's currency. This convention is used throughout this section and in the proof of Lemma 2.2, and all derivatives in this section are evaluated at the balanced baseline unless otherwise indicated.

Equilibrium with balanced trade is $\mathrm{TB}_A(\nu, \tau) = 0$. Linearizing,

$$d\,\mathrm{TB}_A = \mathrm{TB}_{A\nu}\,d\nu + \mathrm{TB}_{A\tau}\,d\tau = 0 \quad \Longrightarrow \quad d\nu^* = -\frac{\mathrm{TB}_{A\tau}}{\mathrm{TB}_{A\nu}}\,d\tau, \tag{2.1}$$

where $\mathrm{TB}_{A\nu} \equiv \partial \mathrm{TB}_A/\partial\nu$ holds the tariff fixed and $\mathrm{TB}_{A\tau} \equiv \partial \mathrm{TB}_A/\partial\tau$ holds ν fixed. At a fixed exchange rate, the tariff produces a positive trade-balance disturbance, $\mathrm{TB}_{A\tau} > 0$, when the induced decline in import payments exceeds any decline in export receipts. Conditional on this disturbance, Eq. (2.1) implies an appreciation if and only if a depreciation improves

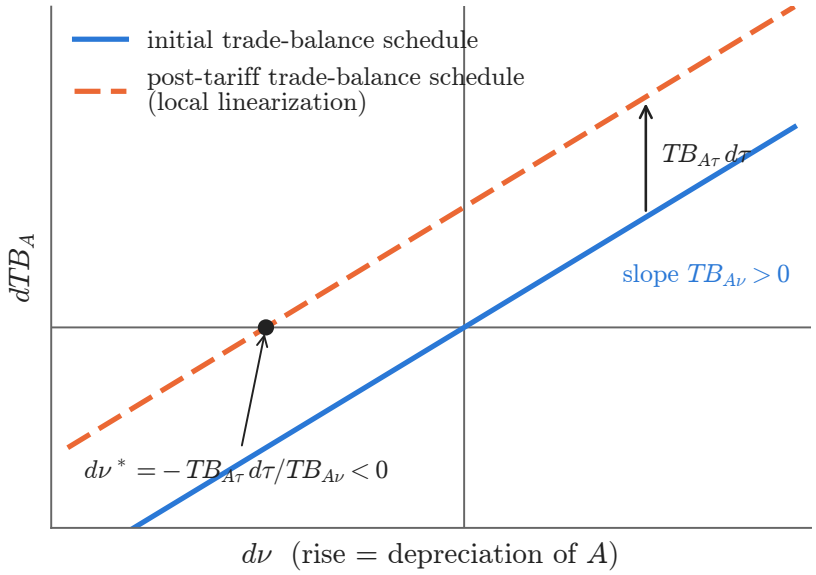


Figure 1: Bilateral tariff adjustment. At a fixed exchange rate, the tariff improves the trade balance; under Marshall–Lerner, equilibrium is restored through appreciation.

the trade balance, $\mathrm{TB}_{A\nu} > 0$.

To express $\mathrm{TB}_{A\nu}$ in elasticities, let Q_X and Q_M be A's export and import quantities, and hold border prices fixed in the producer's currency, as in the nested-CES model, with units chosen so that each border price is one. A's export receipts are then Q_X and its import payments $e_{AB}Q_M$, both in A's currency, so $\mathrm{TB}_A = Q_X - e^{\nu}Q_M$. Define

$$\varepsilon_X \equiv \frac{\partial \log Q_X}{\partial \nu}, \qquad \varepsilon_M \equiv -\frac{\partial \log Q_M}{\partial \nu},$$

at fixed τ and at the balanced baseline. Both are positive when a depreciation raises export volume and lowers import volume. Let $X \equiv Q_X = e_{AB}Q_M$ be the common baseline border value of exports and imports. Differentiating, $\mathrm{TB}_{A\nu} = Q_X\varepsilon_X - e_{AB}Q_M(1-\varepsilon_M) = X(\varepsilon_X + \varepsilon_M - 1)$: the -1 is the valuation effect of a depreciation on the payment for each imported unit. The condition $\mathrm{TB}_{A\nu} > 0$ is therefore the Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$. Figure 1 illustrates; the parallel shift it shows is a first-order approximation around the baseline.

The bilateral appreciation result therefore requires two conditions: the tariff must improve the trade balance at the initial exchange rate, $\mathrm{TB}_{A\tau} > 0$, and a depreciation must improve the trade balance, $\mathrm{TB}_{A\nu} > 0$. Appendix D.1 states sufficient conditions for both. With three or more countries, however, these conditions no longer sign every bilateral exchange-rate response.

2.2 Multilateral adjustment

Now consider N countries, indexed by $\mathcal{I} = \{1, \ldots, N\}$, and let $\mathcal{I}_{-A} = \mathcal{I} \setminus \{A\}$. Take A's currency as numeraire. The $N-1$ bilateral rates e_{Aj}, $j \in \mathcal{I}_{-A}$, span all exchange-rate changes. Trade balances sum to zero (Walras' law), which leaves $N-1$ independent trade-balance conditions, indexed by $i \in \mathcal{I}_{-A}$:

$$\mathrm{TB}_i\Big(\{e_{Aj}\}_{j\in\mathcal{I}_{-A}}, \{\tau_{jk}\}_{(j,k)\in\mathcal{T}}\Big) = 0, \qquad i \in \mathcal{I}_{-A},$$

where τ_{jk} is country j's ad valorem tariff on imports from k and $\mathcal{T} \equiv \{(j,k) \in \mathcal{I}\times\mathcal{I} : j \neq k\}$ is the set of all $N(N-1)$ ordered country pairs. Stack the bilateral rates as $\nu = (\nu_l)_{l\in\mathcal{I}_{-A}}$ with $\nu_l = \log e_{Al}$, in increasing order of l, and the tariffs as $\tau = (\tau_{jk})_{(j,k)\in\mathcal{T}}$, in lexicographic order of (j,k). Linearizing,

$$J\,d\nu + G\,d\tau = 0, \qquad J_{il} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \nu_l}\right|_{\tau}, \qquad G_{i,(jk)} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \tau_{jk}}\right|_{\nu}, \tag{2.2}$$

for $i, l \in \mathcal{I}_{-A}$ and $(j,k) \in \mathcal{T}$, so $J \in \mathbb{R}^{(N-1)\times(N-1)}$ and $G \in \mathbb{R}^{(N-1)\times N(N-1)}$. Consider a tariff change along a fixed direction,

$$d\tau = a\,ds, \qquad a \in \mathbb{R}^{N(N-1)}, \quad \|a\|_2 = 1,$$

where the unit norm serves only to normalize the scalar s. Define $F \equiv Ga$, the marginal vector of trade-balance disturbances per unit increase in s, evaluated at unchanged exchange rates. Substituting into Eq. (2.2),

$$J\,\frac{d\nu}{ds} + Ga = 0.$$

Provided J is nonsingular, therefore,

$$\boxed{\frac{d\nu}{ds} = (-J)^{-1}F.} \tag{2.3}$$

Equation (2.3) separates the trade-balance effects of the tariff at unchanged exchange rates, F, which include expenditure switching, tariff revenue and income effects, from the exchange-rate movements required to restore external balance, $(-J)^{-1}$. In the bilateral case both are scalars, and Eq. (2.3) reduces to Eq. (2.1). With $N \geq 3$, however, the elements of F need not have the same sign: a tariff imposed by A on B can improve a bystander C's trade balance at unchanged exchange rates. The response of any bilateral exchange rate, including ν_C,

therefore depends on the full vector of tariff-induced trade-balance disturbances, not on its own element alone.

2.3 A multilateral Marshall–Lerner condition

Definition 2.1 (Multilateral Marshall–Lerner condition, MML)**.** Let $i, l \in \mathcal{I}_{-A}$. The adjustment mechanism satisfies MML if $J_{il} \geq 0$ for every $l \neq i$, and $(-J)^{-1}$ exists with $(-J)^{-1} \geq 0$ componentwise, that is, $[(-J)^{-1}]_{il} \geq 0$ for all i, l. Equivalently, $-J$ is a nonsingular M-matrix.

Under MML, writing $\vartheta_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0$,

$$\frac{d\nu_i}{ds} = \sum_{j \in \mathcal{I}_{-A}} \vartheta_{ij} F_j, \qquad i \in \mathcal{I}_{-A}. \tag{2.4}$$

Every disturbance enters every bilateral response with a nonnegative weight. Hence, componentwise, if $F \geq 0$ then $d\nu/ds \geq 0$, and if $F \leq 0$ then $d\nu/ds \leq 0$, where a positive $d\nu_i/ds$ is a depreciation of A against i. This is the multilateral counterpart of the two-country result.

A primitive sufficient condition for MML is gross substitutability of trade balances, which is stated in terms of the full $N \times N$ Jacobian with respect to currency values. Let $\chi_l \equiv \log v_l$, $l \in \mathcal{I}$, be the log common-unit value of currency l. Since $e_{Aj} = v_j / v_A$,

$$\nu_j = \chi_j - \chi_A, \qquad j \in \mathcal{I}_{-A}.$$

Define $\widetilde{J}_{il} \equiv \partial \mathrm{TB}_i / \partial \chi_l$ for $i, l \in \mathcal{I}$, at fixed tariffs. If each trade balance is homogeneous of degree one in currency values, Euler's identity gives $\sum_{l \in \mathcal{I}} \widetilde{J}_{il} = \mathrm{TB}_i = 0$ at the balanced baseline, and Walras' law, $\sum_{i \in \mathcal{I}} \mathrm{TB}_i \equiv 0$, gives $\sum_{i \in \mathcal{I}} \widetilde{J}_{il} = 0$ for every $l \in \mathcal{I}$. Because χ_A is fixed by the normalization $v_A = 1$, a change in ν_l is a change in χ_l alone, so the reduced Jacobian of Eq. (2.2) is $J = (\widetilde{J}_{il})_{i,l \in \mathcal{I}_{-A}}$. Trade balances are gross substitutes if

$$\widetilde{J}_{il} \geq 0 \qquad \text{for all } i, l \in \mathcal{I},\ l \neq i,$$

that is, if an appreciation of currency l, a rise in χ_l, weakly improves every other country's trade balance.

Lemma 2.2. *If each trade balance is homogeneous of degree one in currency values, $\widetilde{J}_{il} \geq 0$ for all $i, l \in \mathcal{I}$ with $l \neq i$, the reduced matrix $J = (\widetilde{J}_{il})_{i,l \in \mathcal{I}_{-A}}$ is irreducible, and $\widetilde{J}_{iA} > 0$ for at least one $i \in \mathcal{I}_{-A}$, then MML holds.*

Lemma 2.2 is the trade-balance analogue of the gross-substitutes stability results of Hicks (1939), Mosak (1944), Metzler (1945) and Arrow et al. (1959). MML ensures that multilateral adjustment is well behaved, and a disturbance vector of one sign produces exchange-rate responses of the same sign. When F contains both positive and negative elements, however, MML alone does not determine the sign of an individual bilateral exchange-rate response.

2.4 Symmetric trading partners

Equation (2.3) holds at any balanced baseline. When the tariffing country's partners are symmetric, it can be solved using two eigenspace-specific eigenvalues of $-J$ and two of $G^{(A)}$.

Restrict attention to tariff changes imposed by A, $t = (t_j)_{j \in \mathcal{I}_{-A}}$ with $t_j \equiv d\tau_{Aj}$, and let $G^{(A)} \in \mathbb{R}^{(N-1)\times(N-1)}$ be the corresponding restriction of the tariff-incidence matrix,

$$G^{(A)}_{ij} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \tau_{Aj}}\right|_{\nu}, \qquad i, j \in \mathcal{I}_{-A},$$

so that $J\, d\nu + G^{(A)} t = 0$. Assume that the primitives and the balanced baseline allocation are invariant to every relabeling of A's $N-1$ partners. This implies that, for every $(N-1)\times(N-1)$ partner-permutation matrix P,

$$PJP' = J \qquad \text{and} \qquad PG^{(A)}P' = G^{(A)}.$$

A free-trade or common-external-tariff baseline is an example only when the partner primitives are also symmetric. We then call A's partners symmetric. With $\mathbf{1}$ the $(N-1)$-vector of ones, decompose

$$t = \bar{t}\,\mathbf{1} + \widetilde{t}, \qquad \bar{t} = \tfrac{1}{N-1}\mathbf{1}'t, \qquad \mathbf{1}'\widetilde{t} = 0. \tag{2.5}$$

We call $\bar{t}$ *average protection* and $\widetilde{t}$ *relative treatment*. Any permutation-invariant matrix has the form $aI + b\mathbf{1}\mathbf{1}'$, so the one-dimensional common subspace $\mathrm{span}\{\mathbf{1}\}$ and the $(N-2)$-dimensional deviation subspace $\{v : \mathbf{1}'v = 0\}$ are eigenspaces of both $G^{(A)}$ and $-J$:

$$G^{(A)}\mathbf{1} = g_S\mathbf{1}, \quad G^{(A)}v = g_D v; \qquad (-J)\mathbf{1} = j_S\mathbf{1}, \quad (-J)v = j_D v \qquad (\mathbf{1}'v = 0).$$

The eigenvalues j_D and g_D each have multiplicity $N-2$.

We measure A's effective exchange rate by its nominal effective exchange rate (NEER) with baseline bilateral trade weights. With f_{ij} the border value of country i's imports from

j,

$$w_{Aj} \equiv \frac{f_{Aj} + f_{jA}}{\sum_{k \neq A}(f_{Ak} + f_{kA})}, \qquad \sum_{j \neq A} w_{Aj} = 1,$$
$$E_A \equiv \prod_{j \neq A} e_{Aj}^{w_{Aj}}, \qquad d\nu_A^E \equiv d \log E_A = \sum_{j \neq A} w_{Aj}\, d\nu_j,$$

holding the baseline weights fixed for the local comparative static. This is the effective exchange rate throughout the paper. Under partner symmetry the bilateral trade weights are equal, $w_{Aj} = 1/(N-1)$, so in this section the NEER reduces to the simple average of the bilateral log exchange rates, $d\nu_A^E = \frac{1}{N-1}\sum_{j \neq A} d\nu_j$. For partners $j, k \in \mathcal{I}_{-A}$ the cross rate is $e_{jk} = e_{Ak}/e_{Aj}$, units of j's currency per unit of k's.

Proposition 2.3 (Average protection and relative treatment)**.** *Let partners be symmetric, $N \geq 3$, and $j_S, j_D \neq 0$, so that J is nonsingular. For any tariff vector t,*

$$d\nu = \lambda_{\text{avg}}\, \bar{t}\, \mathbf{1} + \lambda_{\text{rel}}\, \widetilde{t}, \qquad \lambda_{\text{avg}} = \frac{g_S}{j_S}, \quad \lambda_{\text{rel}} = \frac{g_D}{j_D}. \tag{2.6}$$

Hence

$$d\nu_A^E = \lambda_{\text{avg}} \bar{t}, \qquad d\nu_j - d\nu_A^E = \lambda_{\text{rel}} \widetilde{t}_j, \qquad d \log e_{jk} = d\nu_k - d\nu_j = \lambda_{\text{rel}}(t_k - t_j). \tag{2.7}$$

Moreover, under the permutation-symmetry assumptions above, MML is equivalent to $j_S > 0$ and $j_D \geq j_S$.

The two components lie in complementary subspaces, $\mathbb{R}^{N-1} = \text{span}\{\mathbf{1}\} \oplus \{v : \mathbf{1}'v = 0\}$, and each is scaled by its own coefficient. With $N = 3$ these subspaces span a plane, which Fig. 2 uses to illustrate the decomposition.

Proposition 2.3 thus reduces the exchange-rate response to any tariff vector to two coefficients: λ_{avg}, through which average protection moves the common component of A's bilateral rates and hence its effective exchange rate, and λ_{rel}, through which relative treatment moves individual bilateral rates around that component and the cross rates among partners. The decomposition follows from partner symmetry alone and does not rely on the nested-CES structure or any other particular demand system. What remains undetermined are the signs and magnitudes of λ_{avg} and λ_{rel}, and hence the relative strength of the two components. Section 3 introduces a nested-CES model in which these coefficients depend on substitution between domestic and foreign goods and on substitution across foreign suppliers.

3 The Nested-CES Model

This section specializes the decomposition of Section 2.4 into average protection and relative treatment to a nested-CES trade model. In the model, η governs substitution between the domestic tradable and the import composite, and ρ governs substitution across foreign origins. We work locally, to first order around a symmetric, balanced, free-trade benchmark. The central result is that a unilateral tariff always appreciates the tariffing country's NEER and its currency against the target, but depreciates it against untariffed bystanders only when $\rho > N\rho_D$, where ρ_D is a domestic diversion margin defined below. With weak home bias, $\alpha_D \geq 1/N$, a specification with a single elasticity, $\rho = \eta$, cannot produce this reversal.

3.1 Production, preferences and equilibrium

There are $N \geq 2$ countries, $i \in \{1, \ldots, N\}$. Country A is the tariffing country, B is the targeted partner, and lowercase c denotes an individual untariffed bystander; uppercase C is reserved for the composite bystander of Section 3.5. Each country is endowed with L_i units of labor, mobile across sectors, and produces a country-specific tradable and a non-tradable with labor only, $Y_{Ti} = A_{Ti}L_{Ti}$ and $Y_{Ni} = A_{Ni}L_{Ni}$, with $L_{Ti} + L_{Ni} = L_i$. Normalizing each tradable's producer price to one in its own currency gives $w_i = A_{Ti}$ and $P_{Ni} = A_{Ti}/A_{Ni}$. Relative international producer prices therefore move only through exchange rates, and the bilateral exchange rate equals the bilateral terms of trade and the relative wage in a common currency.

Preferences are

$$C_i = C_{Ti}^{\alpha_T} C_{Ni}^{1-\alpha_T}, \quad C_{Ti} = \left[\alpha_D^{1/\eta} C_{Tii}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}, \quad M_i = \left[\textstyle\sum_{j\neq i} \beta_{ij}^{1/\rho} C_{Tji}^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}}, \tag{3.1}$$

with $0 < \alpha_D < 1$, $0 < \alpha_T \leq 1$, $\eta, \rho > 0$ (read as Cobb–Douglas at unit elasticity), $\beta_{ij} > 0$ and $\sum_{j\neq i} \beta_{ij} = 1$. Country i levies an ad valorem tariff τ_{ij} on imports from j, so the consumer price is $p_{ij} = (1+\tau_{ij})e_{ij}$, and the import and tradable price indices are $P_{Mi} = [\sum_{j\neq i} \beta_{ij} p_{ij}^{1-\rho}]^{1/(1-\rho)}$ and $P_{Ti} = [\alpha_D + (1-\alpha_D)P_{Mi}^{1-\eta}]^{1/(1-\eta)}$. Income is

$$I_i = \frac{w_i L_i}{1 - \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}} s_{ij}}, \tag{3.2}$$

where s_{ij} is the tariff-inclusive expenditure share of income on j's good. Tariff revenue is rebated lump-sum and is a domestic transfer, so the external-balance conditions use flows

at border prices:

$$\mathrm{TB}_i = \sum_{k \neq i} f_{ki} - \sum_{j \neq i} f_{ij}, \qquad f_{ij} = \frac{s_{ij}\tilde{I}_i}{1+\tau_{ij}}, \tag{3.3}$$

with $\tilde{I}_i = e_{Ai} I_i$ income in A's currency. Under the normalization $v_A = 1$ of Section 2.1, values in A's currency equal values in the common unit of account, so Eq. (3.3) is the trade-balance measure of Section 2. With A's currency as numeraire, the $N-1$ independent conditions $\mathrm{TB}_i = 0$, $i \neq A$, determine the $N-1$ bilateral rates; wages, prices and expenditure shares follow from the technologies and the demand system Eq. (3.1).

We write $\nu_j = \log e_{Aj}$, with $\nu_A = 0$. The NEER is that of Section 2.4, $d\nu_A^E = \sum_{j \neq A} w_{Aj}\, d\nu_j$ with baseline trade weights $w_{Aj} = (f_{Aj} + f_{jA}) / \sum_{k \neq A}(f_{Ak} + f_{kA})$. The real exchange rate is $q_{Aj} = e_{Aj} P_j / P_A$, with P_i the consumer price index in own currency. A fall in ν_j is an appreciation of A against j and a rise a depreciation; a rise in $\log q_{Aj}$ is a real depreciation of A against j.

The symmetric benchmark is free trade with $\beta_{ij} = 1/(N-1)$, common parameters, and equal endowments and productivities. It retains home bias α_D. Bilateral trade weights then equal $1/(N-1)$, and $m \equiv \alpha_T(1-\alpha_D)$ denotes the import share of income.

3.2 The two-country benchmark

With $N = 2$ there is a single foreign origin, so ρ is inactive. Linearizing gives $d\mathrm{TB}_A = mM_2\, d\nu_B + m\rho_D\, d\tau$, so

$$\left.\frac{d \log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{\rho_D}{M_2},$$
$$\rho_D \equiv \underbrace{\alpha_D \eta}_{\text{domestic substitution}} + \underbrace{(1-\alpha_D)(1-\alpha_T)}_{\text{rebate / non-tradables}}, \qquad M_2 \equiv 1 + 2\alpha_D(\eta - 1) = \varepsilon_X + \varepsilon_M - 1. \tag{3.4}$$

We call ρ_D the domestic diversion margin. A tariff shifts expenditure toward the domestic economy in two ways: through substitution toward the domestic tradable, and through the income effect of the rebated revenue, part of which is spent on non-tradables. An exchange-rate movement raises no revenue and so has no rebate term, which is why ρ_D differs from M_2. With more than two countries, adjustment also separates into a common movement against all foreign suppliers and relative movements across them.

3.3 The N-country case

The following theorem maps the model's primitives into the coefficients of Proposition 2.3.

Theorem 3.1 (Exchange-rate responses in the symmetric nested-CES model)**.** *At the symmetric benchmark define*

$$M_N \equiv \frac{1}{N-1}\Big[1 + \underbrace{N\alpha_D(\eta-1)}_{\textit{domestic–import adjustment}} + \underbrace{(N-2)\rho}_{\textit{foreign-origin adjustment}}\Big]. \tag{3.5}$$

The eigenvalues of Proposition 2.3 are

$$g_S = -\frac{m\rho_D}{N-1}, \qquad g_D = -\frac{m\rho}{N-1}, \qquad j_S = \frac{mM_N}{N-1}, \qquad j_D = Nj_S. \tag{3.6}$$

MML holds if and only if $M_N > 0$, and $\eta \geq 1$ suffices. When it holds,

$$\lambda_{\text{avg}} = -\frac{\rho_D}{M_N}, \qquad \lambda_{\text{rel}} = -\frac{\rho}{NM_N}, \qquad d\nu_j = \lambda_{\text{avg}}\bar{t} + \lambda_{\text{rel}}\widetilde{t}_j. \tag{3.7}$$

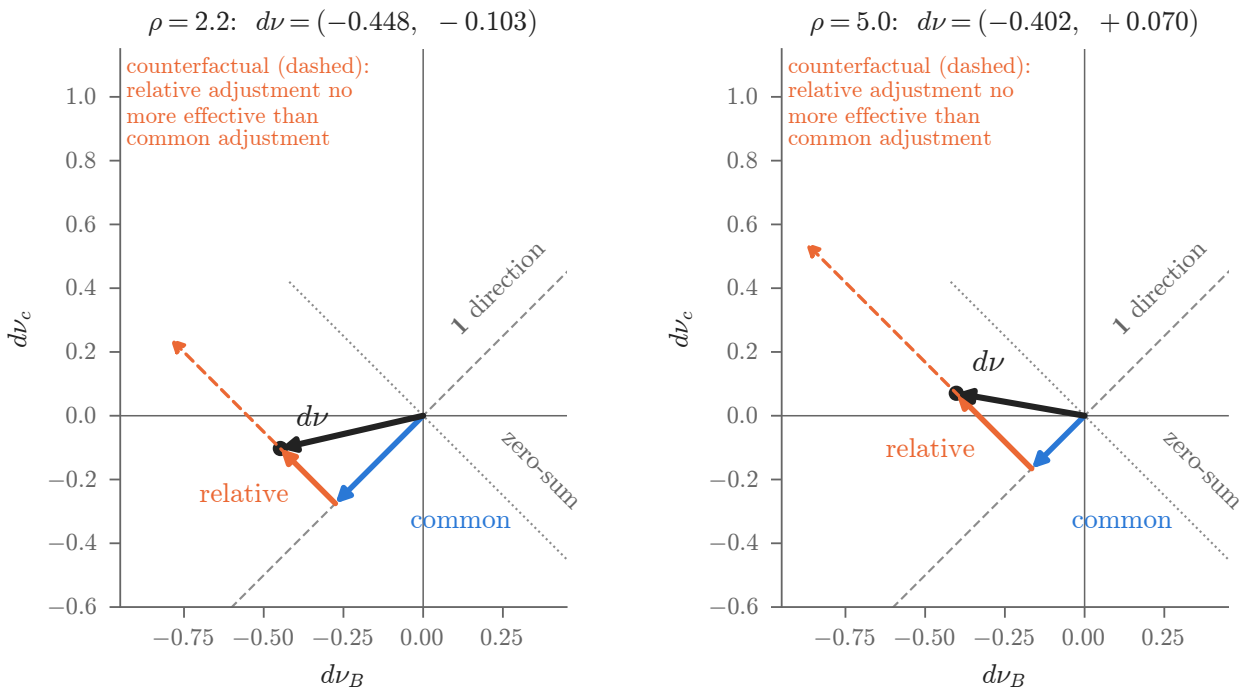
The eigenvalues are derived in Lemma A.1 and Eq. (A.4), and the proof is in Appendix B. M_N is affine in ρ; we write $M_N \equiv M_N(\rho)$ and suppress the argument except in Appendix C.2, where the dependence on ρ matters.

M_N is the sensitivity of A's aggregate trade balance to its effective exchange rate: $d\text{TB}_A = mM_N\, d\nu_A^E$ at the equal weights of the symmetric benchmark. For $N = 2$ it reduces to M_2. For $N > 2$ it includes ρ, because foreign buyers can switch between A's goods and those of other foreign suppliers.

The equality $j_D = Nj_S$ means that zero-sum relative exchange-rate movements are N times as effective as common movements in closing trade imbalances, which is why λ_{rel} carries the factor $1/N$. A common movement shifts only partners' trade with A, whereas a relative movement also shifts partners' trade with *each other*, and at the symmetric benchmark A accounts for a share $1/N$ of the trade on which relative movements act. Because relative treatment is zero-sum, to first order at the symmetric benchmark it leaves tariff revenue, income and the tradable/non-tradable split unchanged, so its disturbance is governed by ρ alone.

3.4 A unilateral tariff

Let A raise its tariff on B alone, $t = d\tau\, e_B$. Its mean component is $\bar{t} = d\tau/(N-1)$, B's relative component is $\widetilde{t}_B = \frac{N-2}{N-1}d\tau$, and each bystander's relative component is $\widetilde{t}_c = -\frac{d\tau}{N-1}$.



(a) $\rho = 2.2$: $\rho_D < \rho < 3\rho_D$ **(b)** $\rho = 5$: $\rho > 3\rho_D$

Figure 2: Average protection and relative treatment. Unilateral tariff with $N = 3$: (a) $\rho_D < \rho < 3\rho_D$; (b) $\rho > 3\rho_D$. *Notes:* $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$.

Proposition 3.2 (Unilateral tariff)**.** *At the symmetric benchmark under MML, to first order,*

$$\frac{d\nu_A^E}{d\tau} = -\frac{\rho_D}{(N-1)M_N} < 0, \qquad \frac{d\nu_B}{d\tau} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N} < 0, \qquad \frac{d\nu_c}{d\tau} = \frac{\rho - N\rho_D}{N(N-1)M_N} \tag{3.8}$$

for every untariffed bystander c. Hence:

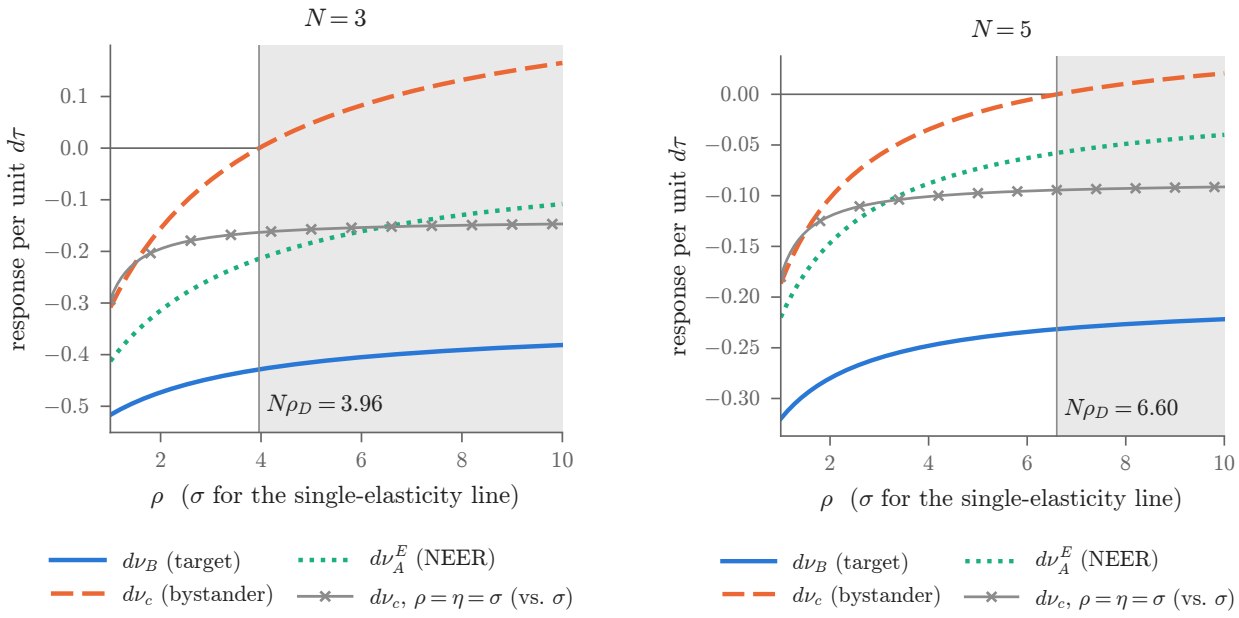
(i) A's NEER appreciates for every N and ρ.

(ii) A appreciates against the target.

(iii) A depreciates against each untariffed bystander if and only if $\rho > N\rho_D$.

(iv) The same threshold applies when A tariffs any $k \leq N-2$ partners. More generally, for a tariff vector with $\bar{t} > 0$, A depreciates against a partner j receiving a below-average tariff increase, $t_j < \bar{t}$, if and only if $t_j/\bar{t} < 1 - N\rho_D/\rho$, which requires $\rho > N\rho_D$.

Against the target, average protection and relative treatment push in the same direction. Against a bystander they are offsetting pressures: average protection pushes toward an appreciation of A, while the bystander's favorable relative treatment pushes toward a depreciation.



(a) $N = 3$ **(b)** $N = 5$

Figure 3: Exchange-rate responses to a unilateral tariff. The bystander response changes sign at $\rho = N\rho_D$. *Notes:* US-like calibration, $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$.

Corollary 3.3 (No reversal with a single elasticity under weak home bias)**.** *Suppose weak home bias, $\alpha_D \geq 1/N$. Then*

$$N\rho_D - \eta = \eta(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) \geq 0, \tag{3.9}$$

so $\rho > \eta$ is necessary for a bystander depreciation. In any specification with a single elasticity across all tradable varieties ($\rho = \eta = \sigma$), $\rho - N\rho_D = -[\sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T)] \leq 0$, so wherever MML holds $d\nu_c/d\tau \leq 0$: A never depreciates against a bystander. The response is strictly negative except at the knife edge $\alpha_D = 1/N$, $\alpha_T = 1$, where it is zero for every $\sigma > 0$. More generally, at $\alpha_D = 1/N$ the response is $-(1-\alpha_T)/(N\sigma)$: if $\alpha_T < 1$ it approaches zero from below as $\sigma \to \infty$, and if $\alpha_T = 1$ it is identically zero.

Under weak home bias, the reversal therefore requires cross-origin substitution to exceed home–import substitution. Evidence in Feenstra et al. (2018) and Li et al. (2022) supports allowing ρ and η to differ. At an illustrative US-like calibration ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, so $\rho_D = 1.32$), the threshold is $N\rho_D = 3.96$ for $N = 3$ and 6.60 for $N = 5$.

At unchanged exchange rates, the trade balance of an individual bystander improves once $\rho > \rho_D$:

$$F_c = \frac{m(\rho - \rho_D)}{(N-1)^2}. \tag{3.10}$$

This is not sufficient for A to depreciate against the bystander in equilibrium. Under MML the exchange-rate response depends on the full vector of trade-balance disturbances, including the target's negative disturbance, and reverses only under the stronger condition $\rho > N\rho_D$. Thus, for $\rho_D < \rho < N\rho_D$, the bystander gains at the initial exchange rates even though A ultimately appreciates against it (Fig. 2a). The gap between the two thresholds reflects the greater effectiveness of relative exchange-rate adjustment identified in Theorem 3.1: if relative and common movements were equally effective, the bystander's equilibrium response would instead change sign when its own disturbance does. Figure 3 plots the three responses against ρ.

3.5 The role of the bystander bloc

The bystander threshold $N\rho_D$ rises with the number of countries. To see why, note that when the bystanders are symmetric and a tariff shock treats them alike, no relative price within the bystander bloc moves, so the bloc can be collapsed into a single region.

Proposition 3.4 (Three-region representation)**.** *At the symmetric benchmark with $N \geq 3$, consider a tariff shock that treats the $N-2$ bystanders symmetrically and leaves tariffs among them unchanged, and suppose $M_N \neq 0$ (MML is sufficient). To first order, the response of ν_B, the common response of each bystander, and A's NEER are exactly reproduced by a three-region economy consisting of A, B and a composite bystander C with income $N-2$, home share of tradables $h_C = \alpha_D + (1-\alpha_D)\frac{N-3}{N-1}$, and home–import elasticity*

$$\boxed{\eta_C = \frac{(N-3)\rho + 2\alpha_D\,\eta}{N-3+2\alpha_D}.} \tag{3.11}$$

The complete mapping of trade weights and tariff shocks is given in Appendix B.

The representation shows that the increase in the bystander threshold with N is not a mechanical consequence of combining more countries into a larger bloc. As the number of bystanders rises, a larger share of the composite bloc's activity consists of trade among the original bystanders. That trade adjusts at the cross-origin elasticity ρ, which changes the bloc's internal adjustment and makes relative exchange-rate movements more effective.

This internal trade structure explains why the unilateral threshold rises to $N\rho_D$. The complete mapping and a counterfactual that holds the composite bloc's internal elasticity fixed are reported in Appendix B.

A nominal bystander reversal need not imply a real reversal because the tariff also raises A's consumer price index. At the US-like calibration with $N = 3$, the nominal threshold

is $3\rho_D = 3.96$, while the real threshold is 4.93. Appendix C.2 derives the real-exchange-rate results and the conditions under which the real effective exchange rate continues to appreciate.

The nested-CES model therefore gives economic content to the decomposition in Section 2. Average protection operates through the domestic diversion margin ρ_D, while relative treatment operates through cross-origin substitution ρ. Under a unilateral tariff, the two forces reinforce each other against the target but oppose each other against untariffed bystanders. A bystander depreciation requires $\rho > N\rho_D$, while the effective exchange rate continues to appreciate. We next examine how retaliation changes this balance.

4 Retaliation and Relative Treatment

How does retaliation change the balance between average protection and relative treatment? Under a unilateral tariff, average protection pushes A's currency toward appreciation, while the favorable relative treatment of bystanders pushes in the opposite direction. Retaliation adds a second relative-treatment effect in the bystanders' favor without adding an equivalent average-protection effect.

Because the analysis is local, the exchange-rate responses to simultaneous tariff changes can be obtained by superposition. The main result is that retaliation lowers the bystander-depreciation threshold from $\rho > N\rho_D$ under a unilateral tariff to $\rho > \rho_D$.

4.1 Bilateral retaliation and multilateral trade wars

Proposition 4.1 (Retaliation and trade wars)**.** *At the symmetric benchmark with $N \geq 3$ under MML, to first order:*

(i) Under bilateral retaliation (A tariffs B and B tariffs A by $d\tau$), $d\nu_B = 0$ and

$$\frac{d\nu_c}{d\tau} = \frac{\rho - \rho_D}{(N-1)M_N}, \qquad \frac{d\nu_A^E}{d\tau} = \frac{N-2}{N-1} \cdot \frac{\rho - \rho_D}{(N-1)M_N}. \tag{4.1}$$

Both belligerents depreciate against every bystander, and in effective terms, if and only if $\rho > \rho_D$. Cross rates among bystanders are unchanged.

(ii) Under a multilateral trade war (A tariffs every partner, and every partner retaliates against A), $d\nu_j = d\nu_A^E = \frac{(N-2)(\rho-\rho_D)}{(N-1)M_N}\, d\tau$ for all j, so A depreciates against every partner if and only if $\rho > \rho_D$.

Under bilateral retaliation, the two average-protection effects on the belligerents' bilateral rate cancel, leaving that rate unchanged. Against a bystander, however, the relative-

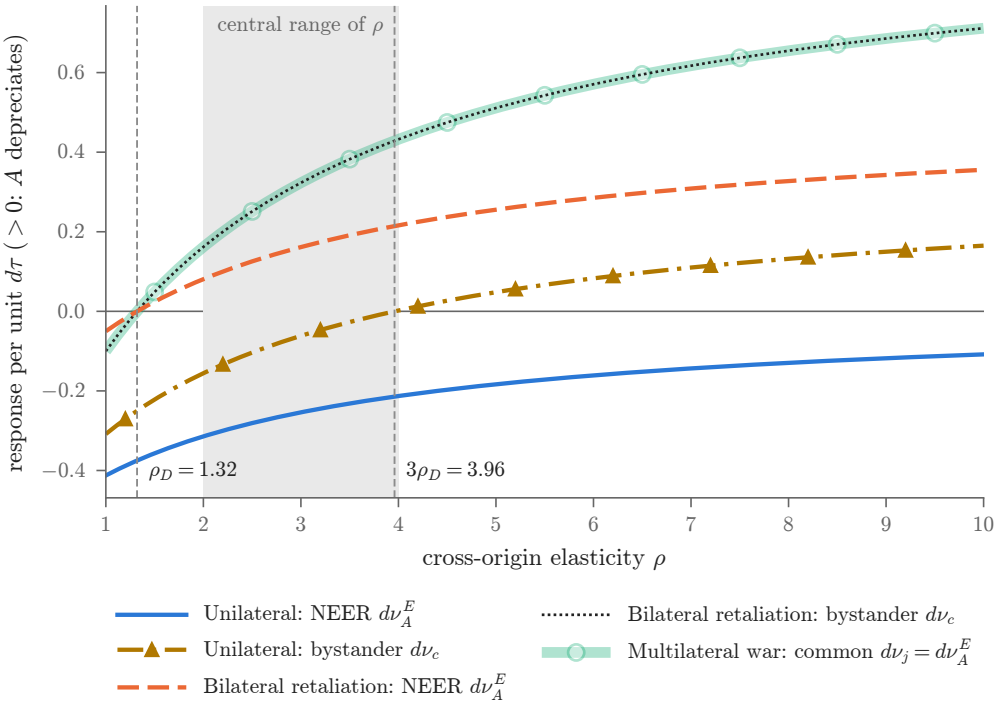


Figure 4: Tariff configurations and exchange-rate responses. Retaliation lowers the bystander-depreciation threshold from $3\rho_D$ to ρ_D when $N = 3$. *Notes:* US-like calibration; shading denotes $\rho \in [2, 4]$.

treatment effects reinforce each other. The bystander threshold therefore falls from $N\rho_D$ under a unilateral tariff to ρ_D, the point at which the bystander's trade balance improves at unchanged exchange rates. When $\rho > \rho_D$, both belligerents depreciate against the bystanders and in effective terms. At the US-like calibration, $\rho_D = 1.32$, below the illustrative range $\rho \in [2, 4]$ shown in Fig. 4.

A multilateral trade war produces the same threshold but shifts the location of adjustment. Under bilateral retaliation, the belligerents' bilateral rate is unchanged and adjustment occurs against the bystanders. In a multilateral trade war there are no untreated bystanders, so all of A's bilateral rates move together. In both cases, retaliation increases the importance of relative treatment relative to average protection.

4.2 Connection to preferential trade

The bystander response is also a third-country terms-of-trade effect of discriminatory protection, connecting the analysis to the preferential-trade literature. In the canonical competing-exporter model, a tariff on one exporter improves the terms of trade of an untariffed exporter because demand is diverted toward it. In the present model, the tariffing country also produces a competing tradable, so diverted expenditure can instead shift toward domestic

production.

The bystander therefore gains only when cross-origin substitution is sufficiently strong relative to domestic diversion. The formal comparison with the competing-exporter model, the extension to a positive common external tariff, and the results for discrete preferential cuts are reported in Appendix C.1.

5 Asymmetry

Partner symmetry makes average protection and relative treatment separate cleanly: the first moves the effective exchange rate, while the second moves bilateral rates around it. Asymmetry in country sizes, trade shares, or preferences breaks this exact decomposition. This section asks which conclusions survive. The central distinction is between uniform and discriminatory protection. Uniform protection preserves the conventional local terms-of-trade and welfare result under general asymmetry, while discriminatory protection can overturn it when the target is small or lightly weighted and substitution toward larger bystanders is sufficiently strong.

Three exchange-rate objects must be distinguished. Bilateral rates describe incidence across individual partners. The trade-weighted NEER summarizes the overall nominal exchange-rate movement. The import-weighted aggregate terms of trade is the welfare-relevant object in the model. These objects coincide at the symmetric benchmark but can differ under asymmetry.

5.1 Aggregate terms of trade and welfare

Let b_{Aj} denote partner j's share in A's baseline import expenditure, and let $R_j^B \equiv \partial \nu_j / \partial \tau_{AB}$ be the response of A's bilateral rate against j to a marginal tariff on B. Define

$$R_{A|B} \equiv \sum_{j \neq A} b_{Aj} R_j^B. \tag{5.1}$$

Because producer prices are fixed in their own currencies, a tariff on B improves A's import-weighted aggregate terms of trade if and only if $R_{A|B} < 0$.

The welfare result is stated at a balanced free-trade baseline of the model of Section 3.1 with interior shares, lump-sum rebate of tariff revenue, and exchange rates differentiable in tariffs, which holds whenever J is nonsingular.

Lemma 5.1 (First-order welfare)**.** *Under these assumptions, let $d\tau$ be any marginal tariff vector, imposed by A or by other countries. To first order, the change in the welfare of A's*

representative household, in units of A's income, is

$$\frac{dU_A}{U'_A\, I_A} = -\, m_A \sum_{j \neq A} b_{Aj}\, d\nu_j, \tag{5.2}$$

where m_A is A's import share of income. A marginal tariff change therefore raises A's own welfare to first order if and only if it improves A's import-weighted aggregate terms of trade.

At free trade, the first-order tariff-revenue gain offsets the direct increase in import prices, leaving the welfare effect of the induced exchange-rate adjustment. The formal proof is in Appendix B. For tariffs imposed by A, write

$$\frac{dU_A}{U'_A\, I_A} = \sum_{j \neq A} g_j\, d\tau_{Aj}, \qquad g_j \equiv -m_A\, R_{A|j}, \tag{5.3}$$

where $R_{A|j}$ is the response Eq. (5.1) to a tariff on j alone, so that g_j is the first-order welfare gain to A from a marginal tariff on j.

5.2 Uniform protection

Proposition 5.2 (Uniform protection)**.** *Consider any balanced free-trade baseline at which MML and the assumptions of Lemma 5.1 hold; country sizes, origin shares, home biases and parameters may differ across countries. Let A impose a marginal uniform tariff on all foreign partners, $d\tau_{Aj} = dt > 0$ for all $j \neq A$, with all other tariffs unchanged. Then:*

(i) the tariff disturbs every partner's trade balance by

$$F_j = -f_{Aj}\, \rho_D^A\, dt, \qquad \rho_D^A \equiv \eta_A h_A + (1 - h_A)(1 - \alpha_T^A) > 0, \tag{5.4}$$

where h_A is A's home share of tradable spending and η_A, α_T^A are A's own parameters;

(ii) A's currency appreciates weakly against every partner, $d\nu_j \leq 0$, and strictly against every partner if A imports from all of them or J is irreducible;

(iii) A's import-weighted aggregate terms of trade improve and A's own welfare rises to first order: $\sum_{j \neq A} g_j > 0$.

Uniform protection leaves relative import prices across foreign suppliers unchanged. Cross-origin substitution therefore creates no differential incidence across partners, and the tariff produces a disturbance of the same sign for every partner. Under MML, A's currency appreciates against every partner, its aggregate terms of trade improve, and its own welfare rises to first order.

This is a local result at a balanced free-trade baseline. It does not apply to finite tariffs, retaliation, global welfare, or the globally optimal tariff. Its role here is to establish that asymmetry alone does not overturn the conventional aggregate result. The possibility of reversal arises from discriminatory protection.

5.3 Discriminatory protection

A discriminatory tariff can produce a different aggregate result because the favorable movement against the target and the adverse movements against bystanders receive different weights. Separate the target B from the remaining suppliers and define their import-weighted average response as

$$\bar{R}^B_{-B} \equiv \sum_{j \neq A,B} \frac{b_{Aj}}{1 - b_{AB}} R^B_j.$$

Then

$$R_{A|B} = b_{AB} R^B_B + (1 - b_{AB}) \bar{R}^B_{-B}.$$

Definition 5.3 (Aggregate terms-of-trade condition)**.** Unilateral tariffs by A have the conventional aggregate terms-of-trade effect if, for every target B,

$$b_{AB} R^B_B + (1 - b_{AB}) \bar{R}^B_{-B} < 0. \tag{5.5}$$

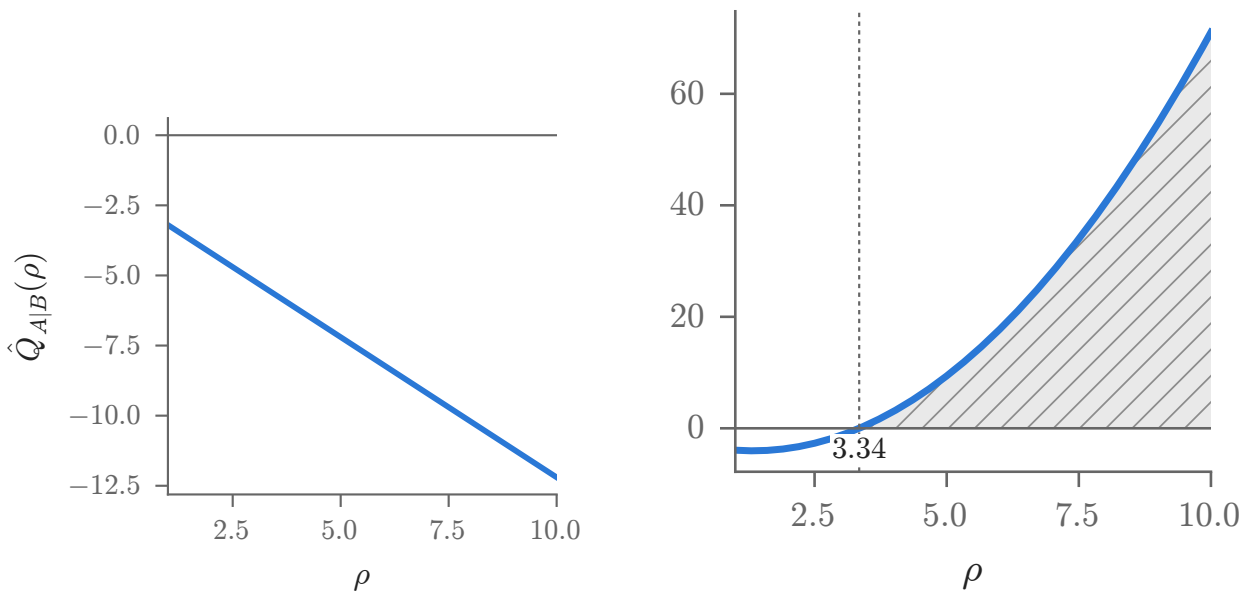
When $R^B_B < 0$ and $\bar{R}^B_{-B} > 0$, the condition fails if and only if

$$\frac{\bar{R}^B_{-B}}{-R^B_B} > \frac{b_{AB}}{1 - b_{AB}}. \tag{5.6}$$

If A appreciates against the target but depreciates on average against the bystanders, the aggregate terms of trade deteriorate only when the adverse bystander movement is sufficiently large relative to the favorable target movement and the bystanders carry sufficient weight in A's import basket. A bilateral bystander reversal is therefore necessary but not sufficient for an aggregate reversal.

In the three-country model, fix a balanced free-trade baseline, that is, the import-origin shares b_{ij}, the home shares h_i, the incomes y_i and all parameters other than ρ, and let only ρ vary. Write the local system as $A(\rho)R = F(\rho)$, with $A \equiv -J$ and $R = (R^B_B, R^B_C)'$. Since ρ enters the linearized share responses (Eq. (A.1)) only through the factor $(1 - \rho)$, every entry of A and F is affine in ρ, and so is the adjugate of the 2×2 matrix A. Hence

$$R_{A|B}(\rho) = \frac{b' \operatorname{adj}(A(\rho)) F(\rho)}{\det A(\rho)}, \qquad b' \operatorname{adj}(A(\rho)) F(\rho) = q_2 \rho^2 + q_1 \rho + q_0 \equiv Q_{A|B}(\rho). \tag{5.7}$$



(a) Symmetric benchmark **(b)** Small, low-weight target

Figure 5: Aggregate terms-of-trade effects under asymmetry. At the symmetric benchmark the conventional aggregate effect always holds. With a small, lightly weighted target, sufficiently strong cross-origin substitution can reverse it.

Under MML $\det A(\rho) > 0$, so at every ρ at which MML holds at the fixed baseline, $R_{A|B}$ has the sign of $Q_{A|B}(\rho)$ and the aggregate condition is $Q_{A|B}(\rho) < 0$.

Proposition 5.4 (Aggregate reversal under discriminatory protection)**.** *Fix a balanced free-trade baseline of the three-country model and consider values of ρ for which MML holds. At the symmetric benchmark, a marginal unilateral tariff improves the tariffing country's aggregate terms of trade for every admissible ρ. Under asymmetry, a marginal discriminatory tariff can instead worsen the aggregate terms of trade and reduce the tariffing country's welfare to first order. Such a reversal is more likely when the target is small, carries a low weight in the tariffing country's imports, and foreign suppliers are sufficiently close substitutes.*

The mechanism is a weighted-incidence effect. An appreciation against a small or lightly weighted target makes only a limited contribution to the import price index. Strong cross-origin substitution redirects expenditure toward larger untariffed suppliers, whose currencies can appreciate against the tariffing country. If these adverse bystander movements are sufficiently large and sufficiently heavily weighted, they dominate the favorable movement against the target.

Figure 5 compares the symmetric benchmark with a small, lightly weighted target, whose parameterization is given in Appendix C.3. In the local model, at the same balanced free-

trade baseline satisfying MML, a marginal uniform tariff raises A's welfare (Proposition 5.2), while a marginal discriminatory tariff can lower it. The two results are compatible because Proposition 5.2 restricts only the sum of the partner-specific welfare gradients in Eq. (5.3), whereas Eq. (5.5) requires each component g_B to be positive. With $N = 3$, $g_B + g_C > 0$, so $g_B < 0$ implies $g_C > 0$.

5.4 Bilateral rates, the NEER, and heterogeneous bystanders

Under asymmetry, bilateral rates, the NEER, and the aggregate terms of trade need not move together. The NEER weights bilateral movements by total trade, including exports, whereas the aggregate terms of trade uses import weights. A favorable movement against the target can therefore receive different weight in the two aggregates, producing different reversal thresholds.

With four or more countries, heterogeneous bystanders generally cannot be collapsed into the exact representative bystander of Proposition 3.4. Their average response depends on the internal adjustment structure of the bystander bloc. The three-country model should therefore be interpreted as a diagnostic of the sign mechanism, not as a quantitative representation of a heterogeneous trade network. Appendix C.3 gives the formal block-matrix representation and numerical examples.

5.5 A stylized reading of the 2025 episode

The 2025 exercise is neither a calibration nor a causal test. The model is static, has flexible prices, and is evaluated around balanced trade, while observed exchange rates are high-frequency asset prices affected by many contemporaneous shocks. We use the model only to classify the sign patterns associated with different tariff configurations. For each phase, we identify the tariff configuration, state the sign pattern associated with that configuration, compare it with the observed exchange-rate movements, and limit the inference to sign consistency.

Observed exchange-rate movements do not identify ρ, ρ_D, or the contribution of the trade mechanism. A movement is described as consistent with the model only when its sign matches the model's sign under the stated tariff configuration and parameter condition. Table 1 and Fig. 6 report the data, with a positive number a depreciation of the dollar; the Federal Reserve's broad dollar index is the closest counterpart of the model's trade-weighted NEER, and the index excluding China isolates the bystanders.

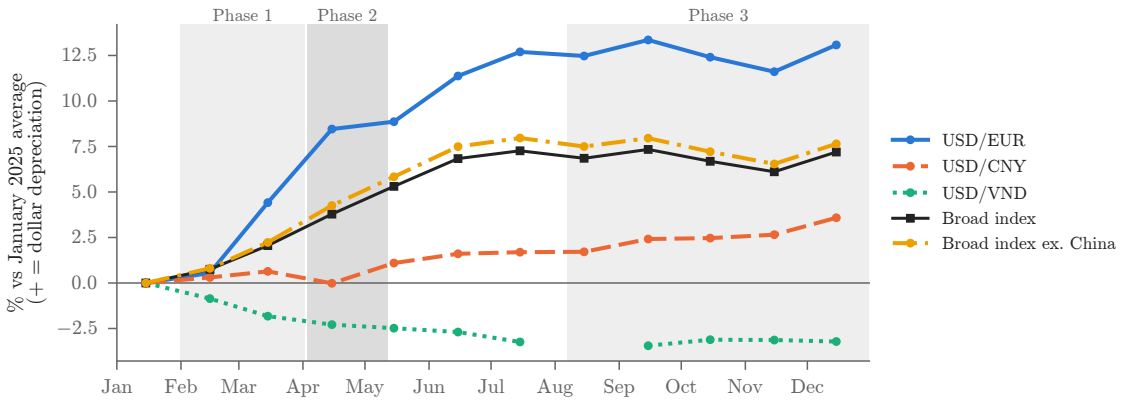
Table 1: Exchange-rate changes within the three phases of the 2025 episode (percent; positive = dollar depreciation)

	Phase 1 Feb 1–Apr 2		Phase 2 Apr 3–May 12		Phase 3 Aug 7–Dec 31	
	Monthly	Daily	Monthly	Daily	Monthly	Daily
EUR	4.4	4.5	3.9	2.2	0.3	0.8
CNY	0.6	−0.3	−0.7	0.9	1.9	2.7
CAD	0.2	1.0	2.7	2.2	−0.7	0.2
MXN	1.6	0.6	1.0	4.5	3.4	3.5
VND	−1.8	−2.1[a]	−0.5	−0.5[a]	0.0	0.5[a]
Broad index	2.1	1.7	1.7	2.3	−0.1	0.6
Broad index ex. China	2.2	1.9	2.0	2.5	−0.3	0.4

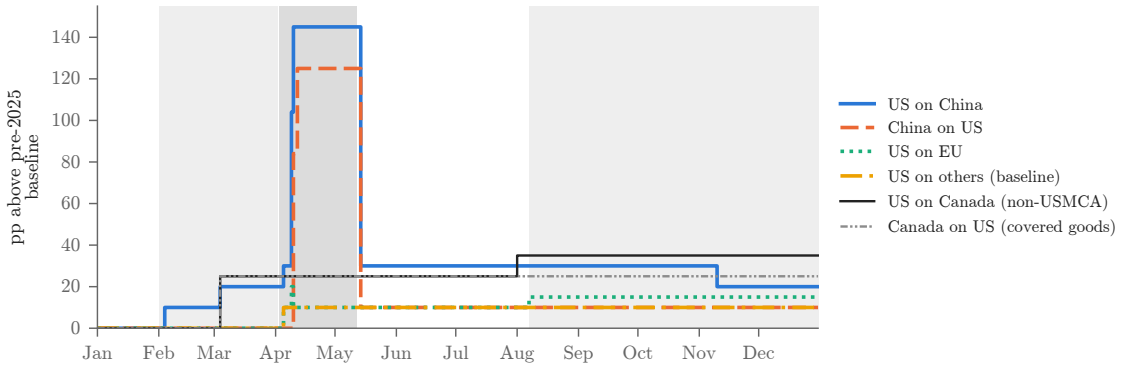
Notes: Monthly columns compare monthly averages (January with March, March with April, July with December); daily columns compare the last available observations at the phase boundaries (January 31 to April 2, April 2 to May 12, August 6 to December 31). VND daily entries (marked [a]) use BIS end-of-month rates. Sources: Federal Reserve H.10 and BIS; see Appendix C.5.

Phase 1: China-focused escalation (February 1–April 2). The United States raised tariffs on China, while China responded with narrower product-specific tariffs. The United States also imposed tariffs on Canada and Mexico, and Canada retaliated, so this phase is not a clean unilateral-tariff experiment. Approximating it as a tariff concentrated on China, Proposition 3.2 associates the configuration with dollar appreciation against the target and, when $\rho > N\rho_D$, depreciation against untariffed bystanders. The dollar was approximately unchanged against the renminbi and weakened against the euro and the broad index excluding China. The bystander movements are consistent with the high-ρ sign pattern, but the mixed tariff configuration and contemporaneous shocks prevent a sharper inference.

Phase 2: US–China retaliation (April 3–May 12). The United States and China raised tariffs sharply against each other, while most other partners faced a common 10 percent US baseline tariff. The configuration is therefore closest to bilateral retaliation with an additional uniform component. Proposition 4.1 associates bilateral retaliation with a stable rate between the belligerents and, when $\rho > \rho_D$, depreciation against bystanders and in effective terms. During the phase, the dollar moved little against the renminbi but depreciated against the euro, the broad index excluding China, and the full broad index. This is the phase most closely aligned with the retaliation sign pattern. The comparison establishes sign consistency only; it does not identify the relevant elasticity or the contribution of the trade mechanism.



(a) Exchange rates, percent deviation from the January 2025 average (positive = dollar depreciation)



(b) Headline tariff increases above the pre-2025 baseline (percentage points)

Figure 6: Exchange rates and headline tariff increases during 2025. Shading denotes the three policy phases. Sources: Federal Reserve, BIS, Federal Register, Department of Finance Canada, and China's State Council Tariff Commission.

Phase 3: Near-uniform tariffs with residual retaliation (August 7–December 31). US tariffs became more nearly uniform across partners, although higher rates and retaliation remained for some countries. The broader uniform component strengthens the average-protection channel, which pushes toward effective dollar appreciation, while the remaining discriminatory and retaliatory components can push in the opposite direction. The broad dollar index was approximately flat within the phase, with no clear effective appreciation. Over 2025 as a whole, the dollar depreciated substantially, by 7.2 percent against the broad index and 7.6 percent against the index excluding China between the January and December monthly averages, with most of that movement occurring before August. The static trade mechanism therefore does not account for the full effective depreciation. A quantitative assessment would require the complete tariff network, substitution elasticities, trade weights, retaliation, and a dynamic mechanism linking expected policy to asset prices.

The comparison across phases is more informative than any individual episode. A con-

figuration dominated by bilateral retaliation is associated with stronger bystander and effective depreciation forces, while broader uniform protection strengthens the average-protection force. The observed movements in Phase 2 have the sign pattern associated with retaliation, whereas Phase 3 provides no corresponding broad appreciation as protection became more uniform. These comparisons illustrate the importance of the tariff configuration without identifying tariffs as the cause of the exchange-rate movements.

6 Conclusion

With two countries, a tariff moves a single exchange rate, so the bilateral exchange rate, the effective exchange rate, and the long-run terms of trade coincide. In a multilateral economy these objects separate. Tariffs change both average protection and relative treatment across foreign suppliers, and the resulting exchange-rate adjustment depends on the full vector of trade-balance disturbances rather than on the tariffing country's bilateral relationship with the target alone.

This distinction changes the conventional tariff result. Under symmetry, a unilateral tariff appreciates the tariffing country's effective exchange rate but can depreciate its currency against untariffed bystanders when cross-origin substitution is sufficiently strong relative to domestic diversion. Retaliation strengthens the relative-treatment force and lowers the bystander threshold. Under asymmetry, uniform protection retains the conventional local aggregate terms-of-trade and welfare result, while discriminatory protection can overturn it when the target is small or lightly weighted and adverse movements against larger bystanders dominate.

The 2025 episode illustrates why the cross-country configuration of protection matters, but the sign comparisons do not identify the trade mechanism or its quantitative contribution. The broader implication is that the effect of tariffs on "the exchange rate" is not well defined without specifying both the exchange-rate object and the configuration of tariffs across trading partners.

References

Amiti, Mary, Stephen J. Redding, and David E. Weinstein, "The Impact of the 2018 Tariffs on Prices and Welfare," *Journal of Economic Perspectives*, 2019, *33* (4), 187–210.

Arrow, Kenneth J. and Leonid Hurwicz, "On the Stability of the Competitive Equilibrium, I," *Econometrica*, 1958, *26* (4), 522–552.

_ , H. D. Block, and Leonid Hurwicz, "On the Stability of the Competitive Equilibrium, II," *Econometrica*, 1959, *27* (1), 82–109.

Bagwell, Kyle and Robert W. Staiger, "An Economic Theory of GATT," *American Economic Review*, 1999, *89* (1), 215–248.

_ and _ , *The Economics of the World Trading System*, Cambridge, MA: MIT Press, 2002.

Barattieri, Alessandro, Matteo Cacciatore, and Fabio Ghironi, "Protectionism and the Business Cycle," *Journal of International Economics*, 2021, *129*, 103417.

Bergin, Paul R. and Giancarlo Corsetti, "The Macroeconomic Stabilization of Tariff Shocks: What Is the Optimal Monetary Response?," *Journal of International Economics*, 2023, *143*, 103758.

Bond, Eric W. and Constantinos Syropoulos, "The Size of Trading Blocs: Market Power and World Welfare Effects," *Journal of International Economics*, 1996, *40* (3–4), 411–437.

Broda, Christian and David E. Weinstein, "Globalization and the Gains from Variety," *Quarterly Journal of Economics*, 2006, *121* (2), 541–585.

Cavallo, Alberto, Gita Gopinath, Brent Neiman, and Jenny Tang, "Tariff Pass-Through at the Border and at the Store: Evidence from US Trade Policy," *American Economic Review: Insights*, 2021, *3* (1), 19–34.

Dornbusch, Rudiger, "Tariffs and Nontraded Goods," *Journal of International Economics*, 1974, *4* (2), 177–185.

Eichengreen, Barry J., "A Dynamic Model of Tariffs, Output and Employment under Flexible Exchange Rates," *Journal of International Economics*, 1981, *11* (3), 341–359.

Erceg, Christopher, Andrea Prestipino, and Andrea Raffo, "Trade Policies and Fiscal Devaluations," *American Economic Journal: Macroeconomics*, 2023, *15* (4), 104–140.

Fajgelbaum, Pablo D., Pinelopi K. Goldberg, Patrick J. Kennedy, and Amit K. Khandelwal, "The Return to Protectionism," *Quarterly Journal of Economics*, 2020, *135* (1), 1–55.

Feenstra, Robert C., Philip Luck, Maurice Obstfeld, and Katheryn N. Russ, "In Search of the Armington Elasticity," *Review of Economics and Statistics*, 2018, *100* (1), 135–150.

Furceri, Davide, Swarnali A. Hannan, Jonathan D. Ostry, and Andrew K. Rose, "Macroeconomic Consequences of Tariffs," IMF Working Paper 19/9, International Monetary Fund 2019.

Hicks, John R., *Value and Capital: An Inquiry into Some Fundamental Principles of Economic Theory*, Oxford: Clarendon Press, 1939.

Itskhoki, Oleg and Dmitry Mukhin, "The Optimal Macro Tariff," NBER Working Paper 33839, National Bureau of Economic Research 2025.

Jeanne, Olivier and Jeongwon Son, "To What Extent Are Tariffs Offset by Exchange Rates?," *Journal of International Money and Finance*, 2024, *142*, 103015.

Kalemli-Özcan, Şebnem, Can Soylu, and Muhammed A. Yıldırım, "Global Trade, Tariff Uncertainty, and the US Dollar," *AEA Papers and Proceedings*, 2026, *116*, 47–52. Longer version: "Tariff Uncertainty and the U.S. Dollar," NBER Working Paper 34728, 2026; also CEPR Discussion Paper 21062.

Krugman, Paul, "The Macroeconomics of Protection with a Floating Exchange Rate," *Carnegie-Rochester Conference Series on Public Policy*, 1982, *16*, 141–182.

Lerner, Abba P., "The Symmetry between Import and Export Taxes," *Economica*, 1936, *3* (11), 306–313.

Li, Wei, Bin Qiu, Zi Wang, Yanbo Zeng, and Yuan Zi, "Armington Elasticities and the Third-Country Effects of Trade Wars," CEPR Discussion Paper 17409, Centre for Economic Policy Research 2022.

Lindé, Jesper and Andrea Pescatori, "The Macroeconomic Effects of Trade Tariffs: Revisiting the Lerner Symmetry Result," *Journal of International Money and Finance*, 2019, *95*, 52–69.

Matveev, Dmitry and Francisco J. Ruge-Murcia, "Tariffs and the Exchange Rate: Evidence from Twitter," Staff Working Paper 2021-36, Bank of Canada 2021. Published in *IMF Economic Review* 72(3), 2024, 1185–1211.

Metzler, Lloyd A., "Stability of Multiple Markets: The Hicks Conditions," *Econometrica*, 1945, *13* (4), 277–292.

_ , "Tariffs, the Terms of Trade, and the Distribution of National Income," *Journal of Political Economy*, 1949, *57* (1), 1–29.

Mosak, Jacob L., *General-Equilibrium Theory in International Trade* number 7. In 'Cowles Commission Monograph.', Bloomington, IN: Principia Press, 1944.

Mundell, Robert A., "Flexible Exchange Rates and Employment Policy," *Canadian Journal of Economics and Political Science*, 1961, *27* (4), 509–517.

_ , "Tariff Preferences and the Terms of Trade," *Manchester School of Economic and Social Studies*, 1964, *32* (1), 1–13.

Ostry, Daniel, Simon Lloyd, and Giancarlo Corsetti, "Trading Blows: The Exchange-Rate Response to Tariffs and Retaliations," Staff Working Paper 1139, Bank of England 2025. Also CEPR Discussion Paper 20452.

Viner, Jacob, *The Customs Union Issue*, New York: Carnegie Endowment for International Peace, 1950.

Werning, Iván, Guido Lorenzoni, and Veronica Guerrieri, "Tariffs as Cost-Push Shocks: Implications for Optimal Monetary Policy," NBER Working Paper 33772, National Bureau of Economic Research 2025.

A The Linearized System

This appendix derives the linear system Eq. (2.2) of the nested-CES model of Section 3.1 and evaluates it at the symmetric benchmark.

A.1 General linearization

All derivatives are first-order and evaluated at a balanced free-trade baseline, $\tau_{ij} = 0$ and $\mathrm{TB}_i = 0$ for every $i \in \mathcal{I}$. Values are in the common unit of account of Section 2.1, with $v_A = 1$. Write $\chi_l = \log v_l$ for $l \in \mathcal{I}$; then $\nu_j = \log e_{Aj} = \chi_j - \chi_A$, $\nu_A \equiv 0$, and $d\log e_{ij} = d\nu_j - d\nu_i$.

The baseline is described by origin shares b_{ij} within i's import bundle ($b_{ii} = 0$, $\sum_{j\neq i} b_{ij} = 1$), home shares h_i of i's tradable spending, and incomes $y_i \equiv \tilde{I}_i$ in the common unit. The share of i's income spent on j's good is $s_{ij} = \alpha_T^i(1-h_i)b_{ij}$, and the border value of i's imports from j is $f_{ij} = s_{ij}y_i$; η_i, ρ_i and α_T^i may differ by country. Log-differentiating the price indices of Section 3.1, the demand system Eq. (3.1), income Eq. (3.2) and the trade balance Eq. (3.3) gives

$$
\begin{aligned}
dp_{ij} &= d\nu_j - d\nu_i + d\tau_{ij}, \qquad dp_{ii} = 0,\\
d\log P_{Mi} &= \textstyle\sum_{j\neq i} b_{ij}\, dp_{ij}, \qquad d\log P_{Ti} = (1-h_i)\, d\log P_{Mi},\\
d\log s_{ii} &= -(1-\eta_i)(1-h_i)\, d\log P_{Mi},\\
d\log s_{ij} &= (1-\rho_i)(dp_{ij} - d\log P_{Mi}) + (1-\eta_i)h_i\, d\log P_{Mi},\\
d\log \tilde{I}_i &= d\nu_i + \textstyle\sum_{j\neq i} s_{ij}\, d\tau_{ij},\\
d\mathrm{TB}_i &= \textstyle\sum_{k\neq i} f_{ki}(d\log s_{ki} + d\log \tilde{I}_k - d\tau_{ki}) - \sum_{j\neq i} f_{ij}(d\log s_{ij} + d\log \tilde{I}_i - d\tau_{ij}),
\end{aligned}
\tag{A.1}
$$

where dp_{ij} is the log change in i's consumer price of j's good in i's currency. At free trade only the direct revenue term survives in $d\log\tilde{I}_i$, and $d\log f_{ij} = d\log s_{ij} + d\log\tilde{I}_i - d\tau_{ij}$. Stacking $d\mathrm{TB}_i$ for $i \in \mathcal{I}_{-A}$ in the order of Eq. (2.2) gives J and G, and hence $G^{(A)}$, at any balanced free-trade baseline. The $N \times N$ Jacobian $\tilde{J}$ with respect to $(\chi_l)_{l\in\mathcal{I}}$ of Section 2.3 follows with each $d\nu_i$ replaced by $d\chi_i$.

A.2 The symmetric benchmark

At the symmetric benchmark the country subscripts on η, ρ and α_T are dropped; $b_{ij} = 1/(N-1)$, $h_i = \alpha_D$, $y_i = 1$, and $f_{ij} = s_{ij} = m/(N-1)$ with $m = \alpha_T(1-\alpha_D)$.

Lemma A.1 (Jacobian at the symmetric baseline)**.** *At the symmetric balanced baseline, $\tilde{J}_{il} = mM_N/(N-1)$ for $i,l \in \mathcal{I}$ with $l \neq i$, and $\tilde{J}_{ii} = -mM_N$. The reduced matrix $J = (\tilde{J}_{il})_{i,l\in\mathcal{I}_{-A}}$ satisfies*

$$-J = j_S\,(NI - \mathbf{1}\mathbf{1}'), \qquad j_S = \frac{mM_N}{N-1}, \qquad j_D = Nj_S. \tag{A.2}$$

Proof. Set $d\tau = 0$ in the $\tilde{J}$ version of Eq. (A.1), so $dp_{ij} = d\chi_j - d\chi_i$ and $d\log\tilde{I}_i = d\chi_i$. With $\Sigma \equiv \sum_{j\in\mathcal{I}} d\chi_j$,

$$d\log P_{Mk} = \frac{\Sigma - N\,d\chi_k}{N-1}, \qquad \frac{\partial \log P_{Mk}}{\partial \chi_l} = \frac{1 - N\delta_{kl}}{N-1}.$$

With equal weights, $\sum_{j\neq i}(dp_{ij} - d\log P_{Mi}) = 0$, so $\sum_{j\neq i} d\log s_{ij} = (N-1)(1-\eta)\alpha_D\, d\log P_{Mi}$ and

$$d\mathrm{TB}_i = \frac{m}{N-1}\Big[\sum_{k\neq i}(d\log s_{ki} + d\chi_k) - (N-1)(1-\eta)\alpha_D\, d\log P_{Mi} - (N-1)\,d\chi_i\Big].$$

Fix $l \neq i$. In $\sum_{k\neq i}\log s_{ki}$, the term $k = l$ contributes $-(1-\eta)\alpha_D$ and each of the $N-2$ terms $k \notin \{i,l\}$ contributes $[-(1-\rho) + (1-\eta)\alpha_D]/(N-1)$, so

$$\frac{\partial}{\partial\chi_l}\sum_{k\neq i}\log s_{ki} = -(1-\rho)\frac{N-2}{N-1} - \frac{(1-\eta)\alpha_D}{N-1}.$$

The export income terms contribute one, the import price-index term $-(1-\eta)\alpha_D$, and the import income term zero. Using $(N-1) - (1-\rho)(N-2) = 1 + (N-2)\rho$,

$$\tilde{J}_{il} = \frac{m}{(N-1)^2}\Big[(N-1) - (1-\rho)(N-2) + N\alpha_D(\eta-1)\Big] = \frac{mM_N}{N-1}.$$

The trade balance is homogeneous of degree one in currency values and zero at the baseline,

so the row sums of $\widetilde{J}$ vanish and $\widetilde{J}_{ii} = -(N-1)\widetilde{J}_{il} = -mM_N$. Deleting row and column A gives diagonal entries $-j_S(N-1)$ and off-diagonal entries j_S, which is Eq. (A.2). Since $\mathbf{1}'\mathbf{1} = N-1$, $-J$ has eigenvalue j_S on span$\{\mathbf{1}\}$ and Nj_S on the zero-sum subspace. □

At $d\nu = 0$, a change $t = (t_j)_{j\in\mathcal{I}_{-A}}$, $t_j = d\tau_{Aj}$, in A's tariffs moves only A's import prices and income, so among A's partners only i's exports to A change:

$$d\mathrm{TB}_i\Big|_{\nu} = f_{Ai}\Big(d\log s_{Ai} + d\log \tilde{I}_A - t_i\Big) = \frac{m}{N-1}\Big[-\rho_D\,\bar{t} - \rho\,(t_i - \bar{t})\Big], \tag{A.3}$$

using $d\log P_{MA} = \bar{t}$, $d\log s_{Ai} = (1-\rho)(t_i - \bar{t}) + (1-\eta)\alpha_D\bar{t}$, $d\log \tilde{I}_A = m\bar{t}$ and $1-(1-\eta)\alpha_D - m = \rho_D$. Equivalently, $G^{(A)} = \frac{m}{N-1}\Big[-\rho I + \frac{\rho-\rho_D}{N-1}\mathbf{1}\mathbf{1}'\Big]$, with eigenvalues

$$g_S = -\frac{m\rho_D}{N-1}, \qquad g_D = -\frac{m\rho}{N-1}, \tag{A.4}$$

on span$\{\mathbf{1}\}$ and on the zero-sum subspace.

A.3 The two-country case

At $N = 2$ the margin $(1-\rho)$ is inactive, $dp_{AB} = d\log P_{MA} = d\nu_B + d\tau$ and $dp_{BA} = d\log P_{MB} = -d\nu_B$, so

$$\begin{aligned} d\log s_{AB} &= (1-\eta)\alpha_D(d\nu_B + d\tau), & d\log s_{BA} &= -(1-\eta)\alpha_D\,d\nu_B,\\ d\log \tilde{I}_A &= m\,d\tau, & d\log \tilde{I}_B &= d\nu_B. \end{aligned}$$

Substituting into $\mathrm{TB}_A = f_{BA} - f_{AB}$,

$$d\mathrm{TB}_A = m\Big[\Big(1 + 2\alpha_D(\eta-1)\Big)d\nu_B + \Big(1 - m + \alpha_D(\eta-1)\Big)d\tau\Big] = mM_2\,d\nu_B + m\rho_D\,d\tau,$$

with $M_2 = 1 + 2\alpha_D(\eta-1)$ and $\rho_D = 1 - m + \alpha_D(\eta-1) = \alpha_D\eta + (1-\alpha_D)(1-\alpha_T)$. Setting $d\mathrm{TB}_A = 0$ gives Eq. (3.4).

For the elasticity form, let $p = p_{AB} = (1+\tau)e_{AB}$. At $\tau = 0$ incomes are fixed in own currency when ν moves, and import spending is $s_{AB}I_A = pQ_M$, so at fixed τ

$$\varepsilon_M = -\frac{\partial\log Q_M}{\partial\log p} = 1 - \frac{\partial\log s_{AB}}{\partial\log p} = 1 + \alpha_D(\eta-1),$$

and by symmetry $\varepsilon_X = 1 + \alpha_D(\eta-1)$. Hence $\varepsilon_X + \varepsilon_M - 1 = M_2$.

A.4 Unilateral and discriminatory tariffs

Here $N \geq 3$, the baseline is the symmetric benchmark, and $M_N \neq 0$; sign statements additionally use MML, $M_N > 0$.

For a unilateral tariff, $t = d\tau\, e_B$, and an individual bystander c, $\bar{t} = d\tau/(N-1)$, $\widetilde{t}_B = \frac{N-2}{N-1} d\tau$ and $\widetilde{t}_c = -\frac{d\tau}{N-1}$. By Eq. (3.7),

$$d\nu_B = \frac{\lambda_{\text{avg}} + (N-2)\lambda_{\text{rel}}}{N-1}\, d\tau = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}\, d\tau,$$
$$d\nu_c = \frac{\lambda_{\text{avg}} - \lambda_{\text{rel}}}{N-1}\, d\tau = \frac{\rho - N\rho_D}{N(N-1)M_N}\, d\tau,$$

while $d\nu_A^E = \lambda_{\text{avg}}\, d\tau/(N-1) = -\rho_D\, d\tau/[(N-1)M_N]$ and $d\log e_{Bc} = d\nu_c - d\nu_B = -\lambda_{\text{rel}}\, d\tau = \rho\, d\tau/(NM_N)$, as in Eq. (3.8). By Eq. (A.3), the target's and each bystander's disturbances per unit $d\tau$ are

$$F_B = -\frac{m[\rho_D + (N-2)\rho]}{(N-1)^2} < 0, \qquad F_c = \frac{m}{N-1}\Big[-\frac{\rho_D}{N-1} + \frac{\rho}{N-1}\Big] = \frac{m(\rho - \rho_D)}{(N-1)^2},$$

the second of which is Eq. (3.10).

For the single-elasticity case, $N\rho_D = N[1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)]$ rearranges to Eq. (3.9). With $\rho = \eta = \sigma$, $d\nu_c/d\tau = -[\sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T)]/[N(N-1)M_N]$. At $\alpha_D = 1/N$, $N\rho_D - \sigma = (N-1)(1-\alpha_T)$ and $M_N = \sigma$, so $d\nu_c/d\tau = -(1-\alpha_T)/(N\sigma)$.

For a general tariff vector with $\bar{t} > 0$, substituting λ_{avg} and λ_{rel} into $d\nu_j = \lambda_{\text{avg}}\bar{t} + \lambda_{\text{rel}}(t_j - \bar{t})$ shows that, under MML, A depreciates against partner j if and only if

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_D\, \bar{t} \qquad \Longleftrightarrow \qquad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_D}{\rho}. \tag{A.5}$$

If A raises its tariff by $d\tau$ on k partners, $1 \leq k \leq N-1$, then $\bar{t} = k\, d\tau/(N-1)$, $\widetilde{t}_B = (N-1-k)\, d\tau/(N-1)$ for every targeted B and $\widetilde{t}_c = -k\, d\tau/(N-1)$ for every untargeted c, and Eq. (3.7) gives

$$\frac{d\log e_{Ac}}{d\tau} = \frac{k(\rho - N\rho_D)}{N(N-1)M_N}, \qquad \frac{d\log e_{AB}}{d\tau} = -\frac{Nk\rho_D + (N-1-k)\rho}{N(N-1)M_N},$$
$$\frac{d\nu_A^E}{d\tau} = -\frac{k\rho_D}{(N-1)M_N}. \tag{A.6}$$

A.5 Real exchange rates

Let $P_i = P_{Ti}^{\alpha_T} P_{Ni}^{1-\alpha_T}$ be i's consumer price index in its own currency. With P_{Ni} fixed and $d\log P_{Ti} = (1-\alpha_D)\, d\log P_{Mi}$ at the symmetric benchmark, $d\log P_i = m\, d\log P_{Mi}$, and from $q_{Aj} = e_{Aj} P_j / P_A$,

$$d\log q_{Aj} = d\nu_j + m\,(d\log P_{Mj} - d\log P_{MA}).$$

Let A impose t and its partners impose none. With equal import weights, $d\log P_{MA} = d\nu_A^E + \bar{t}$; partner j's import prices move by $d\nu_k - d\nu_j$ for $k \notin \{A, j\}$ and by $-d\nu_j$ for A's good, so $d\log P_{Mj} = d\nu_A^E - \frac{N}{N-1} d\nu_j$. Hence

$$d\log q_{Aj} = \left(1 - \frac{mN}{N-1}\right) d\nu_j - m\bar{t}, \qquad d\log q_{Aj} - d\log q_A^E = \left(1 - \frac{mN}{N-1}\right) \lambda_{\text{rel}} \widetilde{t}_j, \tag{A.7}$$

with $d\log q_A^E \equiv \frac{1}{N-1} \sum_{j \neq A} d\log q_{Aj}$. Averaging the first expression and using Eq. (3.7) gives Eq. (C.3). For a unilateral tariff, $\bar{t} = d\tau/(N-1)$, and averaging,

$$\frac{d\log q_A^E}{d\tau} = -\left(1 - \frac{mN}{N-1}\right) \frac{\rho_D}{(N-1) M_N} - \frac{m}{N-1}. \tag{A.8}$$

B Proofs

Notation and the sign convention are those of the main text; $m \equiv \alpha_T(1-\alpha_D) > 0$ at the symmetric benchmark. Unless stated otherwise, derivatives are first-order and evaluated at a balanced free-trade baseline, using the linear system of Appendix A. Appendix C.2 gives the real-exchange-rate extension and its proof.

B.1 Multilateral adjustment and symmetric partners

Proof of Lemma 2.2. Free trade is not required. By homogeneity and Euler's theorem, $\sum_{l \in \mathcal{I}} \widetilde{J}_{il} = \text{TB}_i = 0$ at the balanced baseline. Since $\nu_l = \chi_l - \chi_A$ with χ_A fixed, J is $\widetilde{J}$ with row and column A deleted. Its off-diagonal entries are nonnegative, and for each $i \in \mathcal{I}_{-A}$

$$|J_{ii}| - \sum_{l \in \mathcal{I}_{-A},\, l \neq i} J_{il} = \widetilde{J}_{iA} \geq 0,$$

strictly in at least one row. The diagonal of $-J$ is strictly positive: for $N = 2$, $|J_{BB}| = \widetilde{J}_{BA} > 0$; for $N \geq 3$, $J_{ii} = 0$ would force the nonnegative off-diagonal entries of row i to vanish, contradicting irreducibility. Write $-J = D - K$ with $D = \text{diag}(-J) > 0$ and $K \geq 0$. The matrix $D^{-1}K$ is nonnegative and irreducible, with every row sum at most one and at

least one strictly below one. The spectral radius of an irreducible nonnegative matrix is at most its largest row sum, with equality only if all row sums are equal (Perron–Frobenius), so the spectral radius of $D^{-1}K$ is below one and

$$(-J)^{-1} = (I - D^{-1}K)^{-1}D^{-1} = \sum_{k\geq 0}(D^{-1}K)^k D^{-1} \geq 0.$$

Hence MML holds. Because $D^{-1}K$ is irreducible, the series is entrywise positive, and $\det(-J) = \det D \cdot \det(I - D^{-1}K) > 0$. □

Proof of Proposition 2.3. No demand system, free-trade baseline or stability condition is used. If $PMP' = M$ for every $(N-1)\times(N-1)$ permutation matrix P, conjugating by permutations shows that all diagonal entries are equal and all off-diagonal entries are equal, so $M = aI + b\mathbf{1}\mathbf{1}'$, with eigenvalue $a + (N-1)b$ on $\operatorname{span}\{\mathbf{1}\}$ and a on the zero-sum subspace. For $-J = a_J I + b_J \mathbf{1}\mathbf{1}'$ and $G^{(A)} = a_G I + b_G \mathbf{1}\mathbf{1}'$ this gives $j_S = a_J + (N-1)b_J$, $j_D = a_J$, $g_S = a_G + (N-1)b_G$ and $g_D = a_G$, and for $j_S, j_D \neq 0$

$$(-J)^{-1} = \frac{1}{j_D} I + \frac{1}{N-1}\Big(\frac{1}{j_S} - \frac{1}{j_D}\Big)\mathbf{1}\mathbf{1}'.$$

Thus $(-J)^{-1}$ and $G^{(A)}$ share the two eigenspaces, and applying them to $t = \bar{t}\mathbf{1} + \tilde{t}$ gives Eq. (2.6). Partner symmetry makes the baseline flows $f_{Aj} + f_{jA}$ equal across j, so $w_{Aj} = 1/(N-1)$; since $\tilde{t}$ sums to zero, $d\nu_A^E = \lambda_{\text{avg}}\bar{t}$, and the other two parts of Eq. (2.7) follow directly.

For the MML characterization, the off-diagonal entry of J is $-b_J = (j_D - j_S)/(N-1)$, so the sign pattern holds if and only if $j_D \geq j_S$. If $j_D \geq j_S > 0$, $-J$ is nonsingular and every entry of $(-J)^{-1}$ above is nonnegative, so MML holds. Conversely, under MML the sign pattern gives $j_D \geq j_S$, nonsingularity gives $j_S \neq 0$, and $(-J)^{-1}\mathbf{1} = \mathbf{1}/j_S \geq 0$ gives $j_S > 0$. □

B.2 The symmetric nested-CES model

The baseline is the symmetric benchmark of Section 3.1 with $N \geq 3$; for $N = 2$, Theorem 3.1 reduces to Eq. (3.4).

Proof of Theorem 3.1. The benchmark is invariant to relabeling A's partners, so Proposition 2.3 applies, and the eigenvalues Eq. (3.6) are those of Lemma A.1 and Eq. (A.4). Since $m > 0$, $j_S = mM_N/(N-1) > 0$ if and only if $M_N > 0$, and then $j_D = Nj_S > j_S$. By Proposition 2.3, MML is therefore equivalent to $M_N > 0$; if $M_N \leq 0$, then $j_S \leq 0$ and MML fails. If $\eta \geq 1$, the bracket in Eq. (3.5) is 1 plus a nonnegative term plus $(N-2)\rho > 0$, so

$M_N > 0$. Given $M_N > 0$, dividing Eq. (A.4) by the eigenvalues of $-J$ gives $\lambda_{\text{avg}} = -\rho_D/M_N$ and $\lambda_{\text{rel}} = -\rho/(NM_N)$, and Eq. (3.7) is Eq. (2.6). □

Proof of Proposition 3.2. Equation (3.8) follows from Eq. (3.7) (Appendix A.4). Under MML $M_N > 0$, and $\rho_D > 0$ since $\alpha_D\eta > 0$; hence $d\nu_A^E/d\tau < 0$ and, since $N\rho_D+(N-2)\rho > 0$, $d\nu_B/d\tau < 0$. The denominator of $d\nu_c/d\tau$ is positive, so $d\nu_c/d\tau > 0$ if and only if $\rho > N\rho_D$, with $d\nu_c = 0$ at $\rho = N\rho_D$. For (iv), Eq. (A.6) gives each untargeted partner k times the unilateral response, so the threshold is again $\rho > N\rho_D$. For a general tariff vector with $\bar{t} > 0$, Eq. (A.5) gives the stated condition; when $t_j \geq 0$ its left side is nonnegative, so it can hold only if $1 - N\rho_D/\rho > 0$, that is, $\rho > N\rho_D$, and it then implies $t_j < \bar{t}$. □

Proof of Corollary 3.3. Equation (3.9) is an identity (Appendix A.4); under $\alpha_D \geq 1/N$ both terms on its right are nonnegative, so $N\rho_D \geq \eta$, and since a bystander depreciation requires $\rho > N\rho_D$ (Proposition 3.2), it requires $\rho > \eta$. With $\rho = \eta = \sigma$, Eq. (3.8) gives $d\nu_c/d\tau = (\sigma - N\rho_D)/[N(N-1)M_N]$ with $\sigma - N\rho_D = -[\sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T)] \leq 0$; MML is used only to give $M_N > 0$, so $d\nu_c/d\tau \leq 0$. Since $\sigma > 0$ and $\alpha_D < 1$, the bracket, a sum of two nonnegative terms, vanishes if and only if $\alpha_D = 1/N$ and $\alpha_T = 1$. At $\alpha_D = 1/N$, $M_N = \sigma > 0$, so MML holds for every $\sigma > 0$, and $d\nu_c/d\tau = -(1-\alpha_T)/(N\sigma)$, which is negative and tends to zero as $\sigma \to \infty$ if $\alpha_T < 1$ and is identically zero if $\alpha_T = 1$. □

Proposition 3.4 applies to any marginal tariff change that is invariant to permutations of the $N-2$ bystanders and leaves tariffs among bystanders unchanged: tariffs by A or B common across bystanders, tariffs by all bystanders on A or on B common across bystanders, and tariffs between A and B. The mapping sets $d\tau_{AC} = d\tau_{Ac}$, $d\tau_{CA} = d\tau_{cA}$, $d\tau_{BC} = d\tau_{Bc}$, $d\tau_{CB} = d\tau_{cB}$ (the common values) and retains $d\tau_{AB}, d\tau_{BA}$. In the three-region economy, C has income $y_C = N-2$, home share h_C with $(N-1)h_C = N - 3 + 2\alpha_D$, weight $\frac{1}{2}$ on each of A and B in its import nest, tradable share α_T, cross-origin elasticity ρ and home–import elasticity η_C; A and B place import weight $\frac{1}{N-1}$ on each other and $\frac{N-2}{N-1}$ on C.

Proof of Proposition 3.4. The argument compares the two linearized systems at the benchmark; it does not claim that the nonlinear economies coincide at finite tariffs. MML is sufficient for $M_N \neq 0$ but is not otherwise used.

Common bystander response. By Lemma A.1, $-J = j_S(NI - \mathbf{1}\mathbf{1}')$ with eigenvalues j_S and Nj_S, and $j_S = mM_N/(N-1) \neq 0$, so Eq. (2.2) has a unique solution. Baseline and shock are invariant to permutations of the bystanders, hence so is the solution: $d\nu_c = d\nu_C$ for all c. The equilibrium is then given by $d\text{TB}_B = 0$ and $\sum_c d\text{TB}_c = 0$ in $(d\nu_B, d\nu_C)$.

Flows and A's and B's demand. In the N-country economy $\sum_c f_{Ac} = \sum_c f_{cA} = m\frac{N-2}{N-1}$; in the composite $f_{AC} = m\frac{N-2}{N-1}$ and, since $1 - h_C = 2(1-\alpha_D)/(N-1)$, $f_{CA} = \alpha_T(1-h_C)\frac{1}{2}(N-$

$2) = m\frac{N-2}{N-1}$; likewise for B, and $f_{AB} = f_{BA} = m/(N-1)$ in both, so $s_{CA} = s_{cA} = m/(N-1)$. All bystander goods face the common price change $d\nu_C + d\tau_{Ac}$ in A, so A's import price index, its common share change $d\log s_{Ac}$ and its income response coincide with the composite's, and $\sum_c df_{Ac} = df_{AC}$; the same holds for B.

Intra-bloc trade cancels. Summing Eq. (3.3) over bystanders, the terms $\sum_c \sum_{c' \neq c}(f_{c'c} - f_{cc'})$ cancel identically, so $\sum_c \mathrm{TB}_c$ has the form of $\mathrm{TB}_C = f_{AC} + f_{BC} - f_{CA} - f_{CB}$, and TB_B has the form of the composite's. It remains to show $\sum_c df_{cA} = df_{CA}$ and $\sum_c df_{cB} = df_{CB}$.

Identification of y_C, h_C and η_C. Since $d\log f_{cA} = d\log s_{cA} + d\log \tilde{I}_c - d\tau_{cA}$ is common across c, $\sum_c df_{cA} = f_{CA}\, d\log f_{cA}$, which fixes $y_C = N - 2$. Incomes match because $s_{CA} = s_{cA}$ and $s_{CB} = s_{cB}$, and $d\tau_{cA} = d\tau_{CA}$. For shares, measure prices in bystander currency: because bystander currencies move together and tariffs among bystanders do not change, other bystanders' goods have $dp = 0$. Write $S \equiv dp_{cA} + dp_{cB}$ and $D \equiv dp_{cA} - dp_{cB}$. A bystander's import index moves by $S/(N-1)$, the composite's by $S/2$, so

$$
\begin{aligned}
d\log s_{cA} &= (1-\rho)\frac{D}{2} + \left[(1-\rho)\frac{N-3}{2(N-1)} + (1-\eta)\frac{\alpha_D}{N-1}\right]S, \\
d\log s_{CA} &= (1-\rho)\frac{D}{2} + (1-\eta_C)h_C\frac{S}{2}.
\end{aligned}
$$

The D terms coincide for all (S, D); the S terms coincide if and only if $(1-\eta_C)(N-1)h_C = (1-\rho)(N-3) + 2\alpha_D(1-\eta)$, which, with $(N-1)h_C = N - 3 + 2\alpha_D$, is Eq. (3.11). The argument for B's good replaces D by $-D$, and the residual share matches because shares sum to one.

Restricted Jacobian and solution. Hence the two equilibrium conditions, including tariff terms, are identical to the composite's. Imposing $d\nu_c = d\nu_C$ on row B and on the sum of the bystander rows of $-J$,

$$
-J = j_S \begin{pmatrix} N-1 & -(N-2) \\ -(N-2) & 2(N-2) \end{pmatrix}, \qquad \det(-J) = j_S^2 N(N-2), \tag{B.1}
$$

which for $N \geq 3$ is nonsingular if and only if $j_S \neq 0$, whatever its sign. With disturbances F_B, F_c and $F_C = (N-2)F_c$,

$$
d\nu_C = \frac{F_B + (N-1)F_c}{N j_S}, \qquad d\nu_B = \frac{2F_B + (N-2)F_c}{N j_S}, \tag{B.2}
$$

the unique solution of both systems. The composite's trade weights $\frac{1}{N-1}$ and $\frac{N-2}{N-1}$ reproduce A's NEER at symmetric weights. For $N = 3$, $h_C = \alpha_D$ and $\eta_C = \eta$, so C is the single bystander. □

For the unilateral tariff, $F_B + (N-1)F_c = m(\rho - N\rho_D)/(N-1)^2$ (Appendix A.4), so Eq. (B.2) reproduces the threshold of Proposition 3.2. If instead the composite keeps the mapped y_C, h_C, flows, weights and ρ but replaces η_C by η, it no longer reproduces the N-country economy for $N \geq 4$, and its bystander threshold, computed from the linear system, falls slightly with N instead of rising. The computed thresholds are reported in the online supplement.

B.3 Retaliation, welfare, and asymmetry

Proof of Proposition 4.1. Under MML $M_N > 0$ (Theorem 3.1), and the response is linear in tariffs, so responses to simultaneous tariffs superpose. (i) Shock 1, $d\tau_{AB} = d\tau$, is Eq. (3.8). Shock 2, $d\tau_{BA} = d\tau$, is the same experiment with A and B exchanged, so $d\log e_{BA} = -\frac{N\rho_D+(N-2)\rho}{N(N-1)M_N}d\tau$ and $d\log e_{Bc} = \frac{\rho-N\rho_D}{N(N-1)M_N}d\tau$; with $e_{AB} = 1/e_{BA}$ and $e_{Ac} = e_{AB}e_{Bc}$,

$$d\nu_B^{(2)} = \frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}\,d\tau, \qquad d\nu_c^{(2)} = \frac{\rho}{NM_N}\,d\tau.$$

Adding shock 1, $d\nu_B = 0$ and

$$\frac{d\nu_c}{d\tau} = \frac{\rho - N\rho_D}{N(N-1)M_N} + \frac{\rho}{NM_N} = \frac{\rho - \rho_D}{(N-1)M_N}$$

for every bystander, so cross rates among bystanders are unchanged and $d\log e_{Bc} = d\nu_c - d\nu_B = d\nu_c$. With equal weights, $d\nu_A^E = \frac{N-2}{N-1}d\nu_c$, and B's NEER, $[0 + (N-2)d\nu_c]/(N-1)$, is the same. Since $M_N > 0$ and $N \geq 3$, every response has the sign of $\rho - \rho_D$.

(ii) A's uniform tariff contributes $\lambda_{\text{avg}}d\tau = -\frac{\rho_D}{M_N}d\tau$ to every $d\nu_j$. Partner j's tariff on A is shock 2 with j in the role of B: it contributes $\frac{N\rho_D+(N-2)\rho}{N(N-1)M_N}d\tau$ to $d\nu_j$ and $\frac{\rho}{NM_N}d\tau$ to each of the other $N-2$ partners. Summing,

$$\frac{d\nu_j}{d\tau} = -\frac{\rho_D}{M_N} + \frac{N\rho_D + (N-2)\rho}{N(N-1)M_N} + \frac{(N-2)\rho}{NM_N} = \frac{(N-2)(\rho - \rho_D)}{(N-1)M_N},$$

common to all j, so cross rates among partners are unchanged and $d\nu_A^E$ equals this response, with the sign of $\rho - \rho_D$. □

Proof of Lemma 5.1. Indirect utility is $V_A = \kappa\, I_A/(P_{TA}^{\alpha_T}P_{NA}^{1-\alpha_T})$, so the welfare change in units of income is $d\log I_A - \alpha_T\, d\log P_{TA} - (1-\alpha_T)\, d\log P_{NA}$. By Shephard's lemma, $\alpha_T\, d\log P_{TA} = \sum_j s_{Aj}\, d\log p_{Aj}$. Under the producer-price normalization P_{NA} and the domestic tradable's price are fixed in A's currency, and at free trade $d\log p_{Aj} = d\nu_j + d\tau_{Aj}$. Income is $I_A = w_A L_A/(1 - \sum_j \theta_{Aj}s_{Aj})$ with $\theta_{Aj} = \tau_{Aj}/(1+\tau_{Aj})$; at free trade $\theta_{Aj} = 0$ and $d\theta_{Aj} = d\tau_{Aj}$, so

share changes drop out and $d\log I_A = \sum_j s_{Aj}\, d\tau_{Aj}$. Combining,

$$\frac{dU_A}{U'_A\, I_A} = \sum_j s_{Aj}\, d\tau_{Aj} - \sum_j s_{Aj}(d\nu_j + d\tau_{Aj}) = -m_A \sum_{j\neq A} b_{Aj}\, d\nu_j,$$

with $m_A = \sum_j s_{Aj}$ and $b_{Aj} = s_{Aj}/m_A$. The tariff terms cancel because $\tau_{Aj} = 0$ at the baseline; the cancellation is first order only. Other countries' tariffs enter A's prices and income only through $d\nu$, so the formula holds for any tariff vector. □

Proof of Proposition 5.2. Use Eq. (A.1) with country-specific η_i, ρ_i and α_T^i, so $s_{Aj} = \alpha_T^A(1-h_A)b_{Aj}$ and $m_A = \alpha_T^A(1-h_A)$. At $d\nu = 0$, no country other than A faces a price change, so every flow f_{ij} with $i \neq A$ is unchanged and $F_j = f_{Aj}(d\log s_{Aj} + d\log \tilde{I}_A - dt)$. In A, $dp_{Aj} = dt$ for every j, so $d\log P_{MA} = dt$, the cross-origin term vanishes, $d\log s_{Aj} = (1-\eta_A)h_A\, dt$ and $d\log \tilde{I}_A = m_A\, dt$. Hence

$$F_j = f_{Aj}\Big[(1-\eta_A)h_A + \alpha_T^A(1-h_A) - 1\Big]dt = -f_{Aj}\,\rho_D^A\, dt,$$

which is Eq. (5.4); $\rho_D^A > 0$ since $\eta_A > 0$, $0 < h_A \leq 1$ and $\alpha_T^A \leq 1$, so $F_j \leq 0$, strictly if $f_{Aj} > 0$. Under MML, Eq. (2.4) gives $d\nu_i \leq 0$ for every i. If A imports from every partner, $F < 0$, and each row of the nonsingular nonnegative $(-J)^{-1}$ has a positive entry, so $d\nu_i < 0$ for every i; if J is irreducible, $(-J)^{-1} > 0$ entrywise (Lemma 2.2), so $d\nu_i < 0$ for every i whenever $F \neq 0$. By Lemma 5.1, the welfare change $-m_A \sum_j b_{Aj} d\nu_j$ is nonnegative, and strictly positive in either case since the b_{Aj} sum to one; the aggregate terms of trade improve for the same reason. By linearity this welfare change is $\sum_j g_j\, dt$, so $\sum_j g_j > 0$. □

Proof of Proposition 5.4. By Lemma 5.1, a marginal tariff on B worsens A's aggregate terms of trade, and lowers its welfare to first order, if and only if $R_{A|B} > 0$. At any ρ at which MML holds, $-J$ is a nonsingular M-matrix, so $\det A(\rho) > 0$ and $R_{A|B}$ has the sign of $Q_{A|B}(\rho)$ in Eq. (5.7). At the symmetric benchmark with $N = 3$, Eq. (A.2) gives $A = j_S(3I - \mathbf{1}\mathbf{1}')$ with $j_S = mM_3/2$, so $\det A = 3(mM_3/2)^2$, and with equal import weights $R_{A|B} = d\nu_A^E/d\tau = -\rho_D/(2M_3)$ by Eq. (3.8). Hence $Q_{A|B} = -\frac{3}{8}m^2\rho_D M_3$, which is affine in ρ ($q_2 = 0$) and negative wherever MML holds.

Under asymmetry it suffices to exhibit one baseline. In example (b) of Appendix C.3, a small, lightly weighted target, $Q_{A|B}/q_2 = \rho^2 - 2.6695\rho - 2.2568$ with $q_2 > 0$, and MML holds for $\rho \in [0.01, 200]$. Since $q_0/q_2 < 0$, there is exactly one positive root, $\rho^* = 3.344$, and $R_{A|B} > 0$ for $\rho^* < \rho \leq 200$: the tariff worsens A's aggregate terms of trade and lowers its welfare to first order. The dependence on the target's size and import weight is established numerically in Appendix C.3: $q_2 > 0$ arises when the target is small, has a low import weight,

or both, the root rises as the target becomes larger or more heavily weighted, and a large or heavily weighted target yields no positive root. □

B.4 Verification

The analytical Jacobians and tariff disturbances were checked against centered finite differences of the nonlinear model at symmetric and asymmetric baselines. Closed-form responses were also checked against direct solutions of the linear system, with MML verified at every reported threshold.

C Departures from the Symmetric Free-Trade Benchmark

This appendix extends the local results of Sections 2.4, 3 and 4 to common and finite tariffs, real exchange rates and asymmetric primitives, and documents the 2025 data. Equation (2.3) and the MML condition apply at any balanced baseline. Values not derived analytically are numerical comparative statics computed from the exact model, with the baseline solved from primitives; they are not global theorems. MML holds at every reported root and point. Numerical values use the US-like parameters $(\alpha_D, \alpha_T, \eta) = (0.8, 0.4, 1.5)$, $\rho_D = 1.32$.

C.1 Common and finite tariffs

In the model class of Section 3.1 the bystander's exchange rate is its terms of trade with A, $\mathrm{ToT}_{cA} = e_{Ac}$, so the response in Eq. (3.8) is the third-country terms-of-trade effect of a discriminatory tariff, the central object of the preferential-trade-agreement literature since Viner (1950) and Mundell (1964). In the canonical competing-exporter model, an importer A that does not produce the good buys it from B and C under quasi-linear preferences. Let q_j be exporter j's price, $E_j(q_j)$ its export supply and $M_A(p_A)$ the importer's demand at domestic price p_A, and let A levy a tariff t_B on imports from B. With both exporters active, $p_A = q_B + t_B = q_C$, and market clearing requires $E_B(q_C - t_B) + E_C(q_C) = M_A(q_C)$. Differentiating at $t_B = 0$,

$$\frac{dq_C}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} > 0,$$

so a tariff on B always improves C's terms of trade. In the nested-CES model the bystander gains only if $\rho > N\rho_D$. The difference is that A also produces a competing tradable: part of the demand diverted from B goes to A's own good, along with the rebate's income effect,

so the third-country effect survives only when cross-origin substitution is strong relative to domestic diversion.

Suppose A levies a common tariff τ_0 on all $N-1$ partners, which impose no tariffs of their own. With symmetric partner primitives this baseline is invariant to every relabeling of A's partners, so partner symmetry holds and Proposition 2.3 applies. Tariff changes are measured as $t_j = d\log(1+\tau_{Aj})$. Let $\theta \equiv \tau_0/(1+\tau_0)$, μ_A and μ_P be the import shares of A's and of a partner's tradable expenditure, ω the share of A's good in a partner's import bundle, $\phi \equiv \alpha_T\mu_A/(1-\theta\alpha_T\mu_A)$ the response of A's income to its import expenditure, $X \equiv f_{Aj} = f_{jA}$ the balanced bilateral flow between A and each partner, and $Y \equiv f_{jk}$ the flow between two partners. Linearizing at this baseline gives

$$g_S = -X\rho_D(\tau_0), \qquad g_D = -X\rho, \qquad j_S = XM_S, \qquad j_D = \Big(X + (N-1)Y\Big)M_D, \tag{C.1}$$

where

$$\begin{aligned}
\rho_D(\tau_0) &= 1 - \phi(1-\theta) + (\eta-1)(1-\mu_A)(1+\phi\theta),\\
M_S &= 1 + (\eta-1)(1-\mu_A)(1+\phi\theta) + (\rho-1)(1-\omega) + (\eta-1)(1-\mu_P)\,\omega,\\
M_D &= 1 + (\rho-1)\frac{N-3+\omega}{N-2} + (\eta-1)(1-\mu_P)\frac{N-1-\omega}{N-2}.
\end{aligned}$$

At free trade $\theta = 0$, $\phi = m$, $\mu_A = \mu_P = 1-\alpha_D$ and $\omega = 1/(N-1)$, so $\rho_D(0) = \rho_D$ and $M_S = M_D = M_N$. Tariff disturbances and common exchange-rate movements act only on partners' trade with A, whereas a zero-sum movement v also moves partner j's rate against the other partners, on average by $(N-1)/(N-2)$ times v_j; hence the bases X and $X+(N-1)Y$. With $x \equiv X/(X+(N-1)Y)$, $\lambda_{\text{avg}} = -\rho_D(\tau_0)/M_S$ and $\lambda_{\text{rel}} = -x\rho/M_D$. For a further tariff on B, $t = dt\,e_B$, the bystander response is $d\nu_C = (\lambda_{\text{avg}} - \lambda_{\text{rel}})\,dt/(N-1)$, so A depreciates against C if and only if

$$\rho > \rho^*(\tau_0) \equiv \frac{\rho_D(\tau_0)}{x}\,\frac{M_D}{M_S}. \tag{C.2}$$

At free trade $x = 1/N$, so the factor N in the bystander threshold is $1/x$ at the free-trade baseline. Because ω, μ_P, x and M_S depend on baseline shares that depend on ρ, Eq. (C.2) is implicit in ρ and is solved numerically with the tariff-ridden baseline re-solved at each ρ. By linearity in t, a marginal preferential cut for B, $t = -dt\,e_B$, worsens C's terms of trade under the same condition.

In the computed cases a positive common external tariff raises the threshold above its free-trade value $N\rho_D$, mainly because x falls: the common appreciation of A's currency lowers A's

share of each partner's imports, so relative treatment disturbs a smaller part of the trade on which relative exchange-rate movements act. A concentrated finite tariff, starting from the symmetric free-trade benchmark, works in the opposite direction and lowers the unilateral bystander threshold. The local threshold applies to marginal cuts; in the exact model, a discrete preferential cut that removes the tariff on B entirely requires stronger substitution than the local threshold. The numerical values are reported in the online supplement.

C.2 Real exchange rates

Let P_i be i's consumer price index in its own currency and $q_{Aj} \equiv e_{Aj}P_j/P_A$ the real exchange rate, so that a rise is a real depreciation of A. At the symmetric benchmark the real effective exchange rate is $d\log q_A^E \equiv \frac{1}{N-1}\sum_{j\neq A} d\log q_{Aj}$. With partners imposing no tariffs, to first order (Appendix A.5)

$$d\log q_{Aj} = \Big(1 - \tfrac{mN}{N-1}\Big)d\nu_j - m\bar{t}, \qquad d\log q_A^E = \Big[\Big(1 - \tfrac{mN}{N-1}\Big)\lambda_{\text{avg}} - m\Big]\bar{t}. \tag{C.3}$$

The term $-m\bar{t}$ is the direct effect of the tariff on A's consumer price index, which offsets part of a nominal depreciation.

Proposition C.1 (Real exchange rates)**.** *Consider a marginal unilateral tariff $d\tau > 0$ by A on B at the symmetric benchmark, $N \geq 3$, $m > 0$, partners imposing no tariffs, and define*

$$H_N(\rho) \equiv \Big(1 - \frac{mN}{N-1}\Big)(\rho - N\rho_D) - mN\,M_N(\rho).$$

For every untariffed bystander c, if $M_N(\rho) \neq 0$, $d\log q_{Ac}/d\tau = H_N(\rho)/[N(N-1)M_N(\rho)]$, so wherever MML holds the real bilateral response has the sign of $H_N(\rho)$. Since ρ_D does not depend on ρ, $dH_N/d\rho = 1 - mN$.

(i) If $m < 1/N$, H_N has the unique root

$$\rho_q^*(N) = \frac{[(N-1) - mN]N\rho_D + mN[N\alpha_D(\eta-1)+1]}{(N-1)(1-mN)}.$$

Where MML holds, A appreciates in real terms against c for $\rho < \rho_q^*(N)$ and depreciates for $\rho > \rho_q^*(N)$; if MML holds at $\rho_q^*(N)$, then $\rho_q^*(N) > N\rho_D$.

(ii) If $m = 1/N$, H_N does not depend on ρ: $H_N = -[1+N\alpha_D(\eta-1)+N(N-2)\rho_D]/(N-1)$.

(iii) If $m > 1/N$, H_N is decreasing in ρ, so any crossing has the reverse orientation.

(iv) If $\eta \geq 1$, so that MML holds for every $\rho > 0$, and $m \geq 1/N$, then $H_N(\rho) < 0$ for every

$\rho > 0$*: no positive* ρ *produces a real depreciation against an untariffed bystander.*

(v) Under MML, if $m \leq (N-1)/N$*, A's REER appreciates. Under weak home bias this condition holds automatically, since* $m = \alpha_T(1-\alpha_D) \leq (N-1)/N$*.*

At the US-like calibration and $N = 3$, $m = 0.08 < 1/3$ and $\rho_q^* = 4.93$, against the nominal threshold $3\rho_D = 3.96$.

Proof of Proposition C.1. MML is used only for the sign statements in (i), (iii) and (v) and for the comparison in (i); in (iv) it follows from $\eta \geq 1$. By Eq. (A.7) with $\bar{t} = d\tau/(N-1)$, $d\log q_{Ac}/d\tau = (1 - \frac{mN}{N-1})\, d\nu_c/d\tau - \frac{m}{N-1}$. Substituting $d\nu_c/d\tau$ from Eq. (3.8) and writing $\frac{m}{N-1} = \frac{mN\,M_N}{N(N-1)M_N}$ gives the stated response. Under MML $M_N > 0$, so the response has the sign of H_N. Since M_N in Eq. (3.5) has slope $(N-2)/(N-1)$, H_N is affine in ρ with slope $1 - \frac{mN}{N-1} - \frac{mN(N-2)}{N-1} = 1 - mN$.

(i) For $m < 1/N$ the slope is positive. Multiplying $H_N = 0$ by $N-1$ gives $[(N-1) - mN](\rho - N\rho_D) = mN[1 + N\alpha_D(\eta-1) + (N-2)\rho]$; the coefficient on ρ is $(N-1) - mN - mN(N-2) = (N-1)(1-mN)$, which yields $\rho_q^*(N)$. At the root $[(N-1)-mN](\rho_q^* - N\rho_D) = mN(N-1)M_N(\rho_q^*)$. For $m < 1/N$ the coefficient $(N-1) - mN$ exceeds $N - 2 \geq 1$, and if MML holds at ρ_q^* the right-hand side is strictly positive because $m > 0$ and $M_N(\rho_q^*) > 0$; hence $\rho_q^*(N) > N\rho_D$.

(ii) For $m = 1/N$ the slope is zero, and $H_N = \frac{N-2}{N-1}(\rho - N\rho_D) - \frac{1+N\alpha_D(\eta-1)+(N-2)\rho}{N-1}$, which reduces to the stated constant.

(iii) For $m > 1/N$ the slope is negative.

(iv) If $\eta \geq 1$, every term of M_N in Eq. (3.5) is nonnegative and the first is positive, so MML holds for every $\rho > 0$. Since $(N-1)M_N(0) = 1 + N\alpha_D(\eta-1)$ and $\rho_D = 1 + \alpha_D(\eta-1) - m$,

$$(N-1)H_N(0) = -[(N-1)-mN]N\rho_D - mN[1+N\alpha_D(\eta-1)] = -N\Big[(N-1)(\rho_D - m) + Nm^2\Big].$$

With $\eta \geq 1$, $\rho_D - m \geq 1 - 2m$, so

$$(N-1)(\rho_D - m) + Nm^2 \geq Nm^2 - 2(N-1)m + (N-1) = N\left(m - \frac{N-1}{N}\right)^2 + \frac{N-1}{N} > 0.$$

Hence $H_N(0) < 0$. For $m \geq 1/N$, $dH_N/d\rho \leq 0$, so $H_N(\rho) \leq H_N(0) < 0$ for every $\rho > 0$.

(v) Under MML $M_N > 0$, and $\rho_D > 0$. In Eq. (A.8), the first term is nonpositive if and only if $m \leq (N-1)/N$, and the second, $-m/(N-1)$, is strictly negative because $m > 0$. Under weak home bias, $1 - \alpha_D \leq (N-1)/N$ and $\alpha_T \leq 1$, so $m = \alpha_T(1-\alpha_D) \leq (N-1)/N$. □

C.3 Asymmetric primitives and effective rates

At a general balanced baseline the coefficients of Eq. (A.1) are the baseline shares b_{ij}, the home shares h_i and the incomes y_i. The elasticity ρ enters the share responses only through $(1-\rho)$, so every entry of J and F is affine in ρ, each entry of $\mathrm{adj}(-J)$ is a polynomial of degree at most $N-2$ in ρ, and each numerator $[\mathrm{adj}(-J)F]_i$ a polynomial of degree at most $N-1$. Since $\det(-J) > 0$ under MML, a bilateral response can change sign only at a root of this numerator. At a general asymmetric baseline the $N=3$ numerator is already quadratic in ρ:

$$\operatorname{sign}\frac{d\log e_{AC}}{d\tau} = \operatorname{sign}\left(c_2\rho^2+c_1\rho+c_0\right), \quad c_2 = \alpha_T^2\, b_{AB}b_{AC}b_{CA}b_{CB}(1-h_A)(1-h_C)y_Ay_C, \tag{C.4}$$

where c_1 and c_0 depend on the full pattern of shares and sizes. Provided MML holds, $c_2 > 0$ whenever A and C each import from both foreign origins, so at given shares the response is positive beyond the largest positive root, which equals $3\rho_D$ at the symmetric point. The aggregate numerator $Q_{A|B}$ of Eq. (5.7) has the same structure, and its coefficients (q_2, q_1, q_0) classify the aggregate terms-of-trade condition.

Proposition C.2 (Three cases)**.** *Fix a balanced free-trade baseline of the three-country model as in Section 5.3, and consider values $\rho > 0$ at which MML holds at that baseline. The following three cases are those used in this paper; they are not an exhaustive classification of the signs of (q_2, q_1, q_0).*

(a) Symmetric benchmark: $q_2 = 0$ and $Q_{A|B}(\rho) < 0$ for all ρ. The aggregate condition always holds.

(b) $q_2 > 0$ and $q_0 < 0$: $Q_{A|B}$ has exactly one positive root ρ^*. The condition holds for $\rho < \rho^*$ and fails for $\rho > \rho^*$.

(c) $q_2 < 0$, $q_1 > 0$, $q_0 < 0$ and a positive discriminant: $Q_{A|B}$ has two positive roots $\rho_1 < \rho_2$. The condition fails only for $\rho_1 < \rho < \rho_2$.

Proof of Proposition C.2. At any ρ at which MML holds, $-J$ is a nonsingular M-matrix, so $\det A(\rho) > 0$ and $R_{A|B}$ has the sign of $Q_{A|B}(\rho)$. (a) is shown in the proof of Proposition 5.4. (b) If $q_2 > 0$ and $q_0 < 0$, the product of the roots q_0/q_2 is negative, so there is exactly one positive root ρ^*, with $Q < 0$ on $(0, \rho^*)$ and $Q > 0$ beyond. (c) If $q_2 < 0$, $q_1 > 0$, $q_0 < 0$ and the discriminant is positive, the roots are real with positive sum $-q_1/q_2$ and positive product q_0/q_2, and the downward parabola is positive only between them. The cases are not exhaustive: other sign patterns, such as $q_0 > 0$, or $q_2 < 0$ with no positive root, are possible; in the latter case with $q_0 < 0$, $Q_{A|B} < 0$ and the condition holds at every ρ at which MML

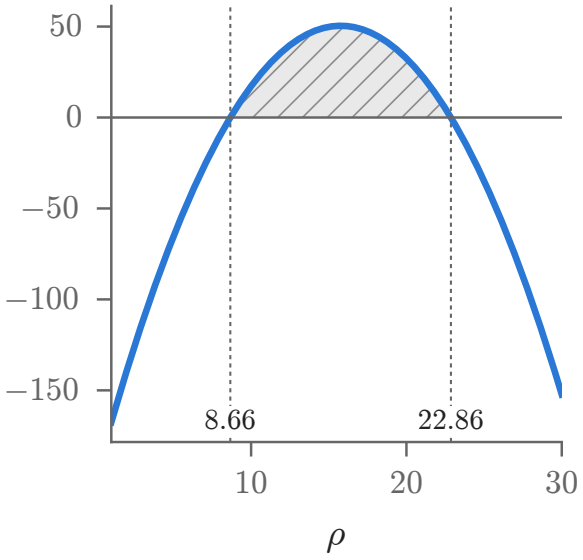


Figure C.1: Aggregate terms-of-trade effects in an asymmetric network with two thresholds, case (c) of Proposition C.2. Shading denotes regions in which a marginal discriminatory tariff worsens the aggregate terms of trade.

holds. □

Case (b) is the economically central asymmetric case, shown in panel (b) of Fig. 5: the conventional aggregate effect holds at low cross-origin substitutability and is reversed above a single threshold. Case (c) requires richer network asymmetry and is reported as a robustness example in Fig. C.1, where the reversal region is non-monotonic in ρ.

The panels of Fig. 5 and Fig. C.1 use the US-like parameters. In (a) and (b) the free-trade baseline is solved from primitives at $\bar{\rho} = 3\rho_D = 3.96$ and then held fixed; $\hat{Q}$ is $Q_{A|B}$ in Eq. (5.7) scaled by $|q_2|$ (by $|q_1|$ when $q_2 = 0$), which preserves its sign and roots, and MML holds at all three baselines for $\rho \in [0.01, 200]$. (a) Symmetric benchmark: $\hat{Q} = -(\rho + 2.2)$. (b) $L = (1, 0.22, 1)$, $\beta_{AB} = 0.2$, $\beta_{AC} = 0.8$, other origin weights symmetric ($b_{AB} = 0.094$): $\hat{Q} = \rho^2 - 2.6695\rho - 2.2568$, with root $\rho^* = 3.344$. (c) A network specified in baseline shares, with $\alpha_T = 0.862$, $\eta = 3.007$, home shares $(0.201, 0.854, 0.360)$, $(b_{AB}, b_{AC}) = (0.525, 0.475)$, $(b_{BA}, b_{BC}) = (0.766, 0.234)$, $(b_{CA}, b_{CB}) = (0.407, 0.593)$ and balancing incomes $(1, 5.126, 0.867)$: $\hat{Q} = -\rho^2 + 31.53\rho - 198.04$, so the condition fails for $8.66 < \rho < 22.86$.

In the family containing example (b), with $L_A = L_C = 1$ and $\beta_{BA} = \beta_{CA} = 0.5$, and with baselines fixed at $\bar{\rho}$, $q_0 < 0$ throughout. A positive q_2, and hence case (b) of Proposition C.2, arises when the target is small, has a low import weight, or both, with an approximate trade-off between the two; the root rises as the target becomes larger or more heavily weighted. For a large or heavily weighted target $q_2 < 0$ and $Q_{A|B}$ has no positive root, so the condition holds at every ρ at which MML holds.

The trade-weighted NEER is $d\nu_A^E = \sum_{j \neq A} w_{Aj}\, d\nu_j$. For $N = 3$ with asymmetric weights, $d\nu_A^E = w_{AB}\, d\nu_B + w_{AC}\, d\nu_C$, so the NEER can depreciate when the bystander reversal occurs and the bystander carries most of A's trade. The bilateral, aggregate terms-of-trade and NEER thresholds are distinct. With $\beta_{AB} = 0.2$, $L_B = 0.223$ (so that $w_{AC} = 0.85$ at $\rho = 3\rho_D$), all other primitives symmetric and the baseline re-solved at each ρ, A depreciates against the bystander for $\rho > 2.59$, its aggregate terms of trade deteriorate ($R_{A|B} > 0$) for $\rho > 3.40$, and its NEER depreciates for $\rho > 3.97$. The NEER threshold is highest because the target's weight in A's trade, $w_{AB} = 0.15$, exceeds its weight in A's imports, $b_{AB} = 0.095$, so the favorable move against the target counts for more in the NEER.

With $N \geq 4$ and bystanders that differ from one another, partition the reduced system into the target B and the set R of the $N-2$ bystanders. Correspondingly, J_{RR} is the block of J giving the bystanders' trade-balance responses to their own rates, J_{RB} the column giving their responses to $d\nu_B$, F_R their tariff disturbances per unit of $d\tau$, and $d\nu_R$ the vector of A's bilateral rates against the bloc. Let $q_j \equiv b_{Aj}/(1 - b_{AB})$, $j \in R$, be bystander j's share in A's imports from the bloc, so that $q'\mathbf{1} = 1$ and $q' d\nu_R/d\tau = \bar{R}^B_{-B}$. The bystanders' rows of the local system read $J_{RB}\, d\nu_B + J_{RR}\, d\nu_R + F_R\, d\tau = 0$. Under MML $-J$ is a nonsingular M-matrix, so its principal block $-J_{RR}$ is nonsingular as well. Solving for $d\nu_R$ and weighting by q gives the bloc's import-weighted response,

$$d\bar{\nu}_R \equiv q' d\nu_R = -q' J_{RR}^{-1} J_{RB}\, d\nu_B - q' J_{RR}^{-1} F_R\, d\tau. \tag{C.5}$$

The identity $R_{A|B} = b_{AB} R^B_B + (1 - b_{AB})\bar{R}^B_{-B}$ always holds, but the bloc's average response depends on J_{RR}, J_{RB} and F_R, not only on the bloc's aggregate size and weight.

C.4 Illustrative U.S.–China–rest-of-world mapping

An illustrative US-China-rest-of-world mapping, including the construction of trade shares, country sizes, origin weights, and the resulting bilateral and aggregate thresholds, is reported in the online supplement. The mapping is illustrative and is not a calibration of exchange-rate magnitudes.

C.5 Data notes for the 2025 episode

Exchange-rate data. Bilateral rates are Federal Reserve H.10 noon buying rates, expressed in dollars per unit of foreign currency (the CNY, CAD and MXN quotes are inverted). The broad index is the H.10 nominal broad dollar index (goods and services), which rises when the dollar appreciates; it is inverted. The index excluding China removes

the renminbi from the daily log change of the broad index using China's 2025 H.10 weight (10.9 percent) and divides by one minus that weight, so the remaining weights are renormalized to sum to one. No daily dong series is available from the H.10 release or from the BIS; the VND row of Table 1 uses BIS monthly averages in the monthly columns and, in the daily columns (marked [a]), BIS end-of-month rates for the same months (end-January to end-March, end-March to end-April, end-July to end-December). Sources: FRED (DEXUSEU, DEXCHUS, DEXCAUS, DEXMXUS, DTWEXBGS), Federal Reserve H.10 trade weights and BIS (VND). "Monthly" compares monthly averages, January with March (Phase 1), March with April (Phase 2) and July with December (Phase 3), windows that do not coincide with the phase dates; "daily" runs from January 31 to April 2, April 2 to May 12 and August 6 to December 31. Panel (a) of Fig. 6 plots monthly averages, at mid-month, relative to each series' January 2025 average; the August dong average is missing in the source.

Tariff data. Panel (b) of Fig. 6 plots statutory headline tariffs above the pre-2025 baseline by effective date: US tariffs on China, which reached 145 percent on April 10 and were reduced to 30 percent on May 14; China's across-the-board tariffs on the United States, which reached 125 percent on April 12; US tariffs on the European Union; the 10 percent US baseline on unlisted partners from April 5; US tariffs on Canada; and Canada's tariffs on covered US goods. Section 232 product tariffs, exclusions and de minimis changes are omitted, and rates are not trade-weighted. US measures are from Federal Register executive orders and Canadian measures from the Department of Finance Canada; Chinese measures are as reported in USDA Foreign Agricultural Service summaries of State Council Tariff Commission announcements. The complete chronology of effective dates and rates is in the online supplement.

Phase definitions. Phase 1 (February 1–April 2) covers the two US tariff increases on China and China's product-specific responses. Phase 2 (April 3–May 12) covers the US–China escalation, the 10 percent baseline of April 5, from which Canada and Mexico were excluded, and the higher country-specific rates of April 9, which were suspended for every partner except China on April 10. Phase 3 (August 7–December 31) begins with the August framework rates.

Classification caveats. The phase classifications are approximate. Tariffs on Canada and Mexico, partial retaliation, exclusions, and contemporaneous policy announcements mean that none of the phases is a clean experiment. Vietnam also shifted between bystander-like and targeted treatment during the year. The online supplement documents these complica-

tions and the complete tariff chronology.

D Further Robustness

This appendix states sufficient conditions for the two-country sign result of Section 2.1. Throughout, e_{Aj} is units of A's currency per unit of j's currency, so a rise is a depreciation of A against j.

D.1 Two-country sufficient conditions

Let $e \equiv e_{AB}$ and let τ be A's tariff on imports from B. Border prices are one in the producer's currency, so with import and export quantities Q_M and Q_X, A's import payments are $M_A = eQ_M$, its export receipts are $X_A = Q_X$, and $\mathrm{TB}_A(e,\tau) \equiv X_A(e,\tau) - M_A(e,\tau)$, where X_A and M_A include all induced effects, including income effects of tariff revenue. If TB_A is continuously differentiable with $\mathrm{TB}_{A,e} \neq 0$ at an equilibrium, the implicit function theorem gives a locally unique $e(\tau)$ with $de/d\tau = -\mathrm{TB}_{A,\tau}/\mathrm{TB}_{A,e}$.

The first condition is $\mathrm{TB}_{A,\tau} > 0$. Because $M_A = eQ_M$ is measured at border prices, import payments fall at fixed e if and only if the import quantity falls. Under a lump-sum rebate of revenue, with ε^H the compensated own-price elasticity of import demand, η_I its income elasticity and θ the revenue share of income,

$$\left.\frac{d\log Q_M}{d\log(1+\tau)}\right|_e = -\frac{\varepsilon^H}{1-\eta_I\theta},$$

so at free trade ($\theta = 0$) the import quantity falls if and only if $\varepsilon^H > 0$. In the nested-CES model at free trade, $\varepsilon^H = \rho_D > 0$, the domestic diversion margin of Eq. (3.4), and $X_{A,\tau} = 0$ because B's income and demand are unchanged at a fixed exchange rate. The second condition is $\mathrm{TB}_{A,e} > 0$, which at a balanced baseline is equivalent to the Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$ (Section 2.1).

When both conditions hold, $de/d\tau < 0$. These are sufficient conditions, not properties of every model: the fiscal closure (revenue wasted, spent by the government or transferred abroad) and the pricing convention (local- or dominant-currency pricing, incomplete pass-through, markups) change the quantity responses and can reverse either sign.

The online supplement also considers sector-specific factors and permanent trade imbalances. These extensions preserve the qualitative mechanism while shifting the quantitative bystander threshold.